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A SPH elastic-viscoplastic model for granular flows and bed-load transport

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A B S T R A C T

An elastic-viscoplastic model (Ulrich, 2013) is combined to a multi-phase SPH formulation (Hu and Adams, 2006; Ghaitanellis et al., 2015) to model granular flows and non-cohesive sediment transport. The soil is treated as a continuum exhibiting a viscoplastic behaviour. Thus, below a critical shear stress (*i.e.* the yield stress), the soil is assumed to behave as an isotropic linear-elastic solid. When the yield stress is exceeded, the soil flows and behaves as a shear-thinning fluid. A liquid-solid transition threshold based on the granular material properties is proposed, so as to make the model free of numerical parameter. The yield stress is obtained from Drucker–Prager criterion that requires an accurate computation of the effective stress in the soil. A novel method is proposed to compute the effective stress in SPH, solving a Laplace equation. The model is applied to a two-dimensional soil collapse (Bui et al., 2008) and a dam break over mobile beds (Spinewine and Zech, 2007). Results are compared with experimental data and a good agreement is obtained.

1. Introduction

The physics of granular material is a major concern for many physical and industrial purposes, because of the wide range of problems it is involved into. In the field of soil mechanics, the study of large deformations and post-failure behaviour of soils are active research topics. In this kind of problem, the soil is assimilated to a continuum and theories of elasticity, plasticity and viscosity are used. In the field of hydraulics, attention is focused on soil-water interactions in order to deal with sediment transport. Usually, the bed evolution is obtained from Exner equation combined to transport formulae involving the empirical Shields erosion criterion. Therefore, the soil is not explicitly modelled and the evolution of the mesh geometry is used to take the bed evolution into account. Although this approach relies on a simple mass balance equation, it has proven its efficiency in modelling bed load transport for large scale problems. However, this approach suffers from lack of physical modelling of the soil behaviour. As a consequence, it is not suitable for problems involving highly dynamic behaviour. Such flows involves several phases and exhibit highly non-linear deformations. They are common in applied hydrodynamics problems and raise many scientific and technical issues. In particular, the modelling of local scour, flow-induced erosion or landslide-induced water waves are particularly complex since they require both an accurate treatment of mechanic behaviour of the soil and of soil-water interactions.

Modelling granular soils is particularly difficult because they behave with some respect to both liquids and solids. Thus, as a first approximation, they can suitably be assimilated to viscoplastic materials (GDR MiDi, 2004; Morichon et al., 2013; Ulrich et al., 2013). Below a critical shear-stress the soil behaves like a solid. When the critical shear stress is exceeded, the soil starts to flow exhibiting a shear thinning behaviour (Campbell, 1990; Jop et al., 2006; GDR MiDi, 2004). Numerically, the solid state is usually approached by a highly viscous state. This kind of regularized viscoplastic model has been applied in the framework of finite volumes (Morichon et al., 2013). However, with such a multi-phase mesh-based method, the presence of a free-surface requires to solve the air-flow and to reconstruct interfaces between air, water and sediment. Moreover, the highly non-linear deformations and the interface fragmentation make these problems difficult to treat with traditional mesh-based Eulerian methods. On the contrary, Lagrangian approaches are particularly adapted for modelling such phenomena. Regularized viscoplastic models have been successfully applied with smoothed particle hydrodynamics method (SPH) to test cases such as saturation driven embankment failure (Ulrich, 2013), submarine landslide (Capone et al., 2010; Xenakis et al., 2015), scouring and sediment flushing problems (Fourtakas and Rogers, 2016; Manenti et al., 2011). Recently Nabian and Farhadi (2016) used a similar approach in the framework of Moving Particle Semi-implicit method (MPS).

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Nevertheless, regularized viscoplastic models exhibit severe drawbacks regarding lowly deformed regions where the soil is supposed to behave like a solid. According to Beverly and Tanner (1992), misleading velocity fields can result from these models when stresses are everywhere very low compared to the yield stress, because they can only mimics the rigid body behaviour in low-stressed regions. This is particularly important regarding erosion and scour development for which an accurate treatment of motionless regions is essential to correctly estimate the bed evolution.

Therefore, in this work the sediment is treated as a continuum whose behaviour law takes account for its granular nature. Ulrich's (2013) elastic-viscoplastic model was thus implemented in an in-house GPU code based on the Cuda language,[1] and improved on physical and numerical aspects. In this model, the sediment behaviour depends on a yield stress determined according to Drucker–Prager's criterion. In unyielded regions, the shear stresses are calculated in line with the linear elastic theory. In yielded regions, a shear thinning rheological law is used and the transitions between solid and liquid states are ensured by a blending function driven by the strain rate magnitude and sediment granular properties.

Three main improvements of Ulrich's model (Ulrich, 2013) were presented; namely a simple and reliable method to compute the effective pressure within the material, which is essential to correctly capture the failure process and the regime transitions, an improvement of solid forces calculation using a SPH second order derivative operator (Espanol and Revenga, 2003), and a physical threshold for the liquid-solid transition. Therefore, contrary to Ulrich's (2013) original model, the mechanical behaviour of the soil does not depend on any numerical parameter in the present model.

The granular and water phases are treated in the frame of Hu and Adams's (2006) SPH multi-phase formulation. The multi-phase model was adapted to semi-analytical wall boundary conditions (Ferrand et al., 2013) in order to model multi-phase flows that both require an accurate pressure and shear stress treatment at the wall, and present complex boundary geometries.

First, the SPH multi-phase model is presented and validated. A description of the soil constitutive equations, as well as their SPH discrete form are then provided. Numerical results are finally compared with experimental data for two 2D cases. We assume that the reader is familiar with SPH (Violeau and Rogers, 2016).

Throughout this document, the sign convention of mechanics is used.

2. Weakly-compressible multi-phase SPH model

2.1. Governing equations

In this work, the sediment is treated as a continuum whose behaviour law takes account for its granular nature. Thus, sediment transport is modelled adopting a multi-phase approach. Water and soil deviatoric stresses are calculated with different behaviour laws, otherwise the same set of equations is used for the two phases. For weakly-compressible flows, the Lagrangian form of continuity equation reads:

$$\frac{d\rho}{dt} = -\rho \operatorname{div} \underline{u} \tag{1}$$

with ρ the density of the material, t the time and $\underline{u}$ the velocity. Then, momentum equation reads:

$$\frac{d\underline{u}}{dt} = -\frac{1}{\rho}\underline{\operatorname{grad}}\, p + \frac{1}{\rho}\underline{\operatorname{div}}\, \underline{\underline{\tau}} + \underline{g} \tag{2}$$

with p the pressure, $\underline{g}$ the gravity and $\underline{\underline{\tau}}$ the shear stress tensor. In the weakly-compressible materials, pressure is related to the density through the following state equation:

$$p = \frac{\rho_0 c_0^2}{\xi}\left[\left(\frac{\rho}{\rho_0}\right)^{\xi} - 1 + p_{bg}^{+}\right] \tag{3}$$

with ρ_0 the reference density of the material, c_0 the numerical speed of sound, ξ the isentropic coefficient and p_{bg}^{+} the dimensionless background pressure. It should be highlighted that the weakly-compressible model is less physical than numerical. The main idea is to allow the density to vary because of the particle Lagrangian motion, while maintaining the compressibility as weak as possible thanks to a large enough speed of sound. Given that it has no physical meaning, the same equation of state can be used for both phases (*i.e.* water and soil), with suitable values of ρ_0 and c_0, and $\xi = 7$.

2.2. SPH multi-phase formulation

2.2.1. Hu and Adams' multi-phase formulation

The main difficulty in simulating multi-phase flows remains in the treatment of discontinuities across phase interfaces. With SPH the density discontinuity raises particular issues. Indeed, the classical SPH differential operators highly depend on the density of neighbouring particles. Consequently, numerical instabilities can occur near the interface for quite large density ratios. Note that the density ratio usually involved in sediment transport problems do not strictly require a multi-phase formulation of SPH operators. As an example, Manenti et al. (2011) and Ulrich (2013) successfully applied one-phase SPH models to such problems. However, Colagrossi and Landrini (2003) showed that the SPH gradient of pressure used in Manenti et al.'s (2011) and Ulrich's (2013) formulations implicitly involves a gradient of density. Then, at the interface of two phases, this formula computes the derivative of a discontinuous field. Even if it does not always lead to numerical instability, this formulation is formally not adapted to multi-phase problem. On the other hand, Fourtakas and Rogers (2016) used Colagrossi and Landrini's (2003) multi-phase formulation. Although this model is suitable for multi-phase problems, the density field is obtained through an explicit integration of the continuity equation. This leads to the accumulation of systematic integration errors, which can make the density and velocity fields become inconsistent (Ferrand et al., 2010; 2013; Vila, 1999). Thus Hu and Adams's (2006) multi-phase formulation will be used in this work.

In standard one-phase SPH formulations, the density ρ of a particle a is usually computed as the interpolation of the density of neighbouring particles b:

$$\rho_a = \sum_{b\in\mathscr{F}} V_b \rho_b w_{ab} \tag{4}$$

with V the particle volume, $w_{ab} = w(r_{ab})$ where w is the SPH kernel (in all this paper, we use Wendland's 5th-order kernel (Wendland, 1995)), $r_{ab} = |\underline{r}_a - \underline{r}_b|$ and $\underline{r}$ the particle position vector. $\mathscr{F}$ denotes the set of free particles. One clearly sees that using Eq. (4) in the vicinity of the interface between two phases of different densities, leads to a numerical smoothing of the interface. This loss of information is a substantial drawback but it can be avoided since it is purely numerical.

To circumvent this issue, Hu and Adams (2006) proposed a formulation based on the interpolation of the inverse volume instead of density:

$$\left(\frac{1}{V}\right)_a = \sum_{b\in\mathscr{F}} w_{ab} \tag{5}$$

Then the definition of density becomes:

$$\rho_a = m_a\left(\frac{1}{V}\right)_a = m_a \sum_{b\in\mathscr{F}} w_{ab} \tag{6}$$

Therefore, the mass m_a of a particle being constant in WCSPH (weakly-

[1] The code is based on a simplified version of the open source GPUSPH solver (GPUSPH official website).

compressible SPH), the variation of density is only due to the evolution of the spatial organisation of neighbouring particles. Thus, the interface smoothing is avoided. In order to simplify notation, we will refer to the particle volume as V_a defined as:

$$V_a = \left[\left(\frac{1}{V} \right)_a \right]^{-1} \tag{7}$$

Hu and Adams (2006) then proposed a pressure gradient that can be derived from (6) and the Lagrangian variational principle of Virtual Work (Grenier et al., 2009):

$$\underline{G}_a\{p_b\} = \frac{1}{V_a} \sum_{b \in \mathscr{F}} (p_a V_a^2 + p_b V_b^2) \underline{\nabla} w_{ab} \approx (\underline{\mathrm{grad}}\, p)_a \tag{8}$$

where < uline > ∇ < /uline > w_{ab} is the kernel gradient at the location of particle a. Note that the pressure gradient does not depend on density any more, and is consequently stable near the interface. In addition, Hu and Adams (2006) also proposed a modified viscous term so as to deal with multiple viscosities, ensuring the continuity of velocity and shear stress across the interface:

$$\begin{aligned} \underline{L}_a\{\eta_b, \underline{u}_b\} &= n_a \sum_{b \in \mathscr{F}} \frac{2\eta_a \eta_b}{\eta_a + \eta_b} (V_a^2 + V_b^2) \frac{\underline{u}_{ab}}{r_{ab}^2} \underline{r}_{ab} \cdot \underline{\nabla} w_{ab} \\ &\approx \left(\underline{\mathrm{div}} \left(\eta\, \underline{\underline{\mathrm{grad}}}\, \underline{u} \right) \right)_a \end{aligned} \tag{9}$$

where η is the dynamic viscosity, $\underline{u}_{ab} = \underline{u}_a - \underline{u}_b$ with $\underline{u}$ the velocity of a particle.

2.2.2. Adaptation to USAW boundary conditions

Hu and Adams' formulation is a straightforward, robust and variationally consistent multi-phase model. However, a rigorous treatment of boundary conditions is necessary to make it capable of simulating flows presenting complex boundary geometries. To do so, the model is adapted to unified semi-analytical wall (USAW) boundary conditions.

USAW boundary conditions framework is based on a mesh used to discretized boundaries of the domain (Ferrand et al., 2013). The mesh is composed of boundary elements ($\mathscr{S}$) being segments in two dimensions and triangles in three dimensions (see Fig. 1a). Free particles are placed at the vertices of the mesh, they move at wall velocity and are referred to as vertex particles ($\mathscr{V}$). Contrary to boundary elements, they carry a mass m and should be taken into account in the continuity equation. Their volume V depends on the local geometry of the boundary and is calculated as a fraction θ of a reference volume $\overline{V}$: $V = \theta \overline{V}$ (see Fig. 1b). In two dimensions, θ is defined as the angle between two connected segments, divided by 2π. Thus, for vertex particles we have $\theta_b \in]0; 1[$. In order to have a general formulation, we also define $\theta_a = 1$ for free particles of matter ($\mathscr{F}$) and $\theta_s = 1/2$ for boundary elements. In three dimensions, θ is calculated in a similar way using solid angles. For boundary elements and vertex particles, the reference volume $\overline{V}$ is obtained from a Shepard interpolation of the inverse volume (for more details, see Ferrand et al., 2013):

$$\forall\, a \in (\mathscr{V} \cup \mathscr{S}) \quad \overline{V}_a = \left[\frac{\sum_{b \in \mathscr{F}} w_{ab}}{\sum_{b \in \mathscr{F}} V_b w_{ab}} \right]^{-1} \tag{10}$$

For free particles, the volume is equal to the reference volume ($\theta = 1$) and is obtained adapting (5) and (7) to this framework:

$$\forall\, a \in \mathscr{F} \quad \overline{V}_a = \left[\frac{1}{\Gamma_a} \sum_{b \in (\mathscr{F} \cup \mathscr{V})} \theta_b w_{ab} \right]^{-1} \tag{11}$$

where Γ is a renormalization factor used to correct the interpolation error due to the lack of particles beyond the boundary:

$$\Gamma_a = \int_{\Omega \cap \Omega_a} w(\underline{r}' - \underline{r}_a) dr' \tag{12}$$

Thus, from (10) and (11), we see that the reference volume is not constant and only depends on the geometrical configuration of neighbouring particles around the point of interest. For free particles, density is obtained adapting Vila's form of the continuity equation (Vila, 1999):

$$\forall\, a \in \mathscr{F} \quad \frac{d(\Gamma_a \rho_a)}{dt} = \frac{d}{dt} \left(m_a \sum_{b \in (\mathscr{F} \cup \mathscr{V})} \theta_b w_{ab} \right) \tag{13}$$

Then, adapting the variationally consistent pressure gradient to USAW boundary conditions, we get an approximation of the pressure gradient:

$$\begin{aligned} \underline{G}_a\{p_b\} = & \frac{1}{\Gamma_a \overline{V}_a} \sum_{b \in (\mathscr{F} \cup \mathscr{V})} \theta_b (p_a \overline{V}_a^2 + p_b \overline{V}_b^2) \underline{\nabla} w_{ab} \\ & - \frac{1}{\Gamma_a \overline{V}_a} \sum_{s \in \mathscr{S}} \frac{1}{\overline{V}_s} (p_a \overline{V}_a^2 + p_s \overline{V}_s^2) \underline{\nabla} \Gamma_{as} \end{aligned} \tag{14}$$

where p is the pressure and $\underline{\nabla} \Gamma_{as}$ is the contribution of segment s to the gradient of Γ_a. For a 1-D segment s, delimited by two vertex particles (v_1, v_2), $\underline{\nabla} \Gamma_{as}$ is defined by:

$$\underline{\nabla} \Gamma_{as} = \left(\int_{v_1}^{v_2} w(r) dl \right) \underline{n}_s \tag{15}$$

where $\underline{n}_s$ is the inward normal of the segment s. Note that the discrete SPH differential operator (14) contains a volumic term, related to free and vertex particles, and a boundary term related to boundary elements. In the absence of boundaries, (14) gives back Hu and Adams' original formula (8).

Similarly, we define the consistent divergence operator for a vector field $\underline{A}$:

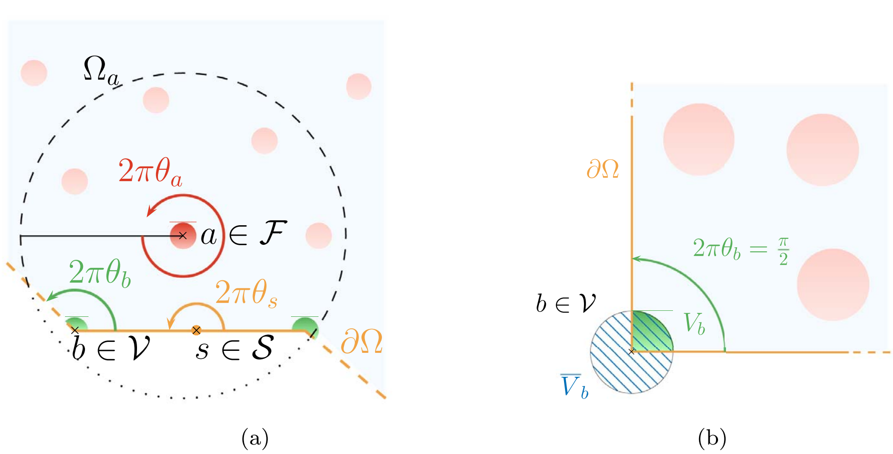

Fig. 1. (a) Sketch of a boundary with: a vertex particle $b \in \mathscr{V}$, θ_v depends on the local shape of the boundary; a segment $s \in \mathscr{S}$ ($\theta_s = 1/2$); a free particle $a \in \mathscr{F}$ ($\theta_a = 1$). (b) Sketch of a vertex particle in a right-angled corner in 2-D, illustrating the relation between volume V_b, the dimensionless angle θ_b and the reference volume $\overline{V}_b$.

$$D_a\{\underline{A}_b\} = -\frac{\overline{V}_a}{\Gamma_a}\sum_{b\in(\mathscr{F}\cup\mathscr{V})}\theta_b\underline{A}_{ab}\cdot\underline{\nabla}w_{ab} + \frac{\overline{V}_a}{\Gamma_a}\sum_{s\in\mathscr{S}}\frac{1}{\overline{V}_s}\underline{A}_{as}\cdot\underline{\nabla}\Gamma_{as} \approx (\text{div}\underline{A})_a \tag{16}$$

where $\underline{A}_{ab}$ denotes $\underline{A}_a - \underline{A}_b$. Now we proceed the same way for the viscous term, paying particular attention to the viscosity of segments and vertex particles. Since we do not want to assign a phase to boundaries, we take $\eta_b = \eta_a$ when . First we adapt Hu and Adams' viscous term (9) to USAW boundary conditions:

$$\underline{L}_a^H\{\eta_b, \underline{u}_b\} = \frac{1}{\Gamma_a V_a}\sum_{b\in(\mathscr{F}\cup\mathscr{V})}\frac{2\eta_a\eta_b}{\eta_a+\eta_b}\theta_b(\overline{V}_a^2+\overline{V}_b^2)\frac{\underline{u}_{ab}}{r_{ab}^2}\underline{r}_{ab}\cdot\underline{\nabla}w_{ab} - \frac{1}{\Gamma_a\overline{V}_a}\sum_{s\in\mathscr{S}}\frac{\eta_a}{\overline{V}_s}(\overline{V}_a^2+\overline{V}_s^2)\frac{\underline{u}_{as}}{r_{as}^2}\underline{r}_{as}\cdot\underline{\nabla}\Gamma_{as} \tag{17}$$

As of now, we will refer to it as Hu and Adams' viscous term. Then, we write the classical USAW Laplacian (resulting from the model of Morris et al., 1997) in the multi-phase framework:

$$\underline{L}_a^M\{\eta_b, \underline{u}_b\} = \frac{1}{\Gamma_a}\sum_{b\in(\mathscr{F}\cup\mathscr{V})}\theta_b\overline{V}_b\frac{4\eta_a\eta_b}{\eta_a+\eta_b}\frac{\underline{u}_{ab}}{r_{ab}^2}\underline{r}_{ab}\cdot\underline{\nabla}w_{ab} - \frac{1}{\Gamma_a}\sum_{s\in\mathscr{S}}2\eta_a\frac{\underline{u}_{as}}{r_{as}^2}\underline{r}_{as}\cdot\underline{\nabla}\Gamma_{as} \tag{18}$$

From now on, we will refer to it as Morris et al.'s (1997) viscous term. Note that, this viscous term also ensures continuity of velocity and shear stress across the interface. The two latter formula are approximations of the Laplacian operators:

$$\underline{L}_a^{H/M}\{\eta_b, \underline{u}_b\} \approx \left(\underline{\text{div}}\left(\eta\,\underline{\underline{\text{grad}}}\;\underline{u}\right)\right)_a \tag{19}$$

A more general discrete second-order operator can also be defined. It will be referred to as Espanol and Revenga's operator and will be denoted $\underline{L}^E$. In the frame of the multi-phase formulation with USAW boundary conditions, it is defined for any vector and scalar fields $\underline{A}$ and B by:

$$\underline{L}_a^E\{B_b, \underline{A}_b\} = \frac{1}{\Gamma_a}\sum_{b\in(\mathscr{F}\cup\mathscr{V})}\theta_b\overline{V}_b\frac{2B_aB_b}{B_a+B_b}((d+2)(\underline{A}_{ab}\cdot\underline{e}_{ab})\underline{e}_{ab} + \underline{A}_{ab})\frac{\underline{\nabla}w_{ab}\cdot\underline{e}_{ab}}{r_{ab}} - \frac{1}{\Gamma_a}\sum_{s\in\mathscr{S}}2B_a\frac{\underline{A}_{as}}{r_{as}^2}\underline{r}_{as}\cdot\underline{\nabla}\Gamma_{as} \tag{20}$$

with d the space dimension. Contrary to (9) and (18), Espanol and Revenga's (2003) formula (20) is not an approximation of the Laplacian. Indeed, Violeau (2009) showed:

$$\underline{L}_a^E\{B_b, \underline{A}_b\} \approx \left(\underline{\text{div}}\left(B\left[\underline{\underline{\text{grad}}}\,\underline{A} + \left(\underline{\underline{\text{grad}}}\,\underline{A}\right)^T\right]\right)\right)_a \tag{21}$$

2.2.3. Rhie and Chow correction

Far from the boundaries and considering the time as a continuous variable, Vila (1999) showed that that the discrete equation of continuity (13) can be related to some kind of implicit time integration of the continuity equation:

$$\frac{d\rho_a}{dt} = -\rho_a D_a\{\underline{u}_b\} \tag{22}$$

An implicit scheme being used for integrating (22) and neglecting the viscous term in integrating the momentum Eq. (2) using the SPH gradient operator (14), we can write:

$$\begin{aligned}\frac{\rho_a^{(n+1)}-\rho_a^{(n)}}{\Delta t} &= -\rho_a^{(n)}D_a\{\underline{u}_b^{(n+1)}\}\\ &= -\rho_a^{(n)}D_a\left\{\underline{u}_b^{(n)} - \frac{\Delta t}{\rho_b^{(n)}}\underline{G}_b\{p_c^{(n)}\} + \Delta t\,\underline{g}\right\}\\ &= -\rho_a^{(n)}(D_a\{\underline{u}_b^{(n)}\} + \Delta tD_a\{\underline{G}_b\{p_c^{(n)}\}\} + \Delta tD_a\{\underline{G}_b\{\underline{g}\cdot\underline{r}_b^{(n)}\}\})\end{aligned} \tag{23}$$

where Δt is the time-step. The latter equation shows that the approximate Laplacian of pressure $D_a\{\underline{G}_b\{p_c^{(n)}\}\}$ is implicitly contained into the time-discretized equation of continuity, acting like a diffusion term. Fatehi and Manzari (2011), and more recently Hashemi et al. (2016), stated that substituting this discrete form of the Laplacian by an SPH Laplacian operator L to be defined can significantly reduce the numerical pressure oscillations. This technique was first introduced for finite volumes by Rhie and Chow (1983). It consists in adding to (13) the difference between $D_a\{\underline{G}_b\{p_c^{(n)}\}\}$ and $L_a\{p_b^{(n)}\}$. This will result in a smoothing term that will tend to zero as the particle spacing δr decreases. Several combinations of discrete Laplacian operators have been tested and we finally chose the following smoothing term:

$$\Delta^{(n)} = \rho_a^{(n)}\Delta t\left[L_a^M\left\{\frac{1}{\rho_b^{(n)}},\ p_b^{(n)}\right\} - D_a^+\left\{\underline{G}_b^-\left\{\frac{p_c^{(n)}}{\rho_b^{(n)}}\right\}\right\}\right] \tag{24}$$

with L^M the Morris operator (18), $\underline{G}^-$ the symmetric SPH gradient defined by:

$$\underline{G}_a^-\{A_b\} = -\frac{\overline{V}_a}{\Gamma_a}\sum_{b\in\mathscr{F}}\theta_b(A_a - A_b)\underline{\nabla}w_{ab} \tag{25}$$

and D^+ the anti-symmetric divergence defined by:

$$D_a^+\{\underline{A}_b\} = \frac{1}{\Gamma_a\overline{V}_a}\sum_{b\in\mathscr{F}}\theta_b(\underline{A}_a\overline{V}_a^2 + \underline{A}_b\overline{V}_b^2)\underline{\nabla}w_{ab} \tag{26}$$

The last two SPH operators have inverse symmetry properties with respect to the original ones used above. This particular combination was not chosen on a theoretical basis but after numerical tests. Moreover, these operators are built on the basis of the multi-phase operators (14) and (16) so the Rhie and Chow's (1983) correction terms are computed from all neighbouring particles, irrespective of the phase they belong to. On the other hand, this correction only occurs between free particles, so the discrete operators in (24)–(26) do not have boundary terms and do not include the vertex particles. The continuity Eq. (13) is corrected as follows:

$$\rho_a^{(n+1)} = \frac{1}{\Gamma^{(n+1)}}\left[\Gamma^{(n)}\rho_a^{(n)} + m_a\sum_{b\in(\mathscr{F}\cup\mathscr{V})}\theta_b(w_{ab}^{(n+1)} - w_{ab}^{(n)})\right] + \Lambda\Delta^{(n)} \tag{27}$$

where Λ is a weighting coefficient. From numerical experiments, $\Lambda = 1$ gives satisfactory results.

Note that no particle shifting algorithm is used in this work.

2.2.4. Time integration scheme

Time integration is done with a fully explicit symplectic method (Ferrand et al., 2013) that leads to the following scheme:

$$\begin{cases}\underline{u}_a^{(n+1)} = \underline{u}_a^{(n)} - \dfrac{\Delta t}{\rho_a^{(n)}}\underline{G}_a\{p_b^{(n)}\} + \dfrac{\Delta t}{\rho_a^{(n)}}\underline{L}_a\{\eta_b, \underline{u}_b^{(n)}\} + \Delta t\,\underline{g}\\ \underline{r}_a^{(n+1)} = \underline{r}_a^{(n)} + \Delta t\,\underline{u}_a^{(n+1)}\\ \rho_a^{(n+1)} = \dfrac{1}{\Gamma^{(n+1)}}\left[\Gamma^{(n)}\rho_a^{(n)} + m_a\displaystyle\sum_{b\in(\mathscr{F}\cup\mathscr{V})}\theta_b(w_{ab}^{(n+1)} - w_{ab}^{(n)})\right] + \Lambda\Delta^{(n)}\end{cases} \tag{28}$$

Two restrictions on the time step must be enforced to ensure the numerical stability of this integration scheme. The first one relies the classical Courant–Friedrichs–Levy (CFL) number. The CFL number compares the time-step to the time for the information to travel the characteristic distance of the simulation (*i.e.* the discretization scale h) at the characteristic velocity of the problem. In WCSPH the CFL number is defined as:

$$C_{\mathrm{CFL}} = \frac{c_0 \Delta t}{h} \tag{29}$$

For the viscous forces, an additional condition must be enforce through the following dimensionless number:

$$C_{\mathrm{visc}} = \frac{\eta \Delta t}{\rho h^2} \tag{30}$$

This number expresses the fact that, the higher the viscosity, the faster the information propagate along the successive layers of the fluid. More details about the numerical stability of WCSPH can be found in Violeau and Leroy (2014) and Hashemi et al. (2016) (the latter includes the effect of density smoothing).

2.2.5. Validation

The present formulation is validated on a two-fluid laminar plane Poiseuille flow in a closed channel, involving two fluids of different kinematic viscosities (ν_1, ν_2) and densities (ρ_1, ρ_2). The 2D flow is driven by a gravity force $\rho \mathrm{g}$ oriented in the direction of the flow $\underline{e}_x$, ($\underline{e}_x$, $\underline{e}_y$) being the base vectors. Periodic open boundaries are used and the height of the channel is $L = 1$ m. The interface between the two fluids is in $y = 0$, the top wall in $y = L/2$ and the bottom wall in $y = -L/2$, where $\underline{e}_y$ denotes the transverse direction. In the present work, the subscript 1 refers to the fluid in the bottom part ($y < 0$). In all simulations, the ratio between δr and the smoothing length h is $\delta rh = 2$. Note that, in this case the Rhie and Chow's (1983) correction presented in Section 2.2.3 was not used. Instead, the Brezzi and Pitkäranta's (1984) correction presented in Ghaitanellis et al. (2015) was used in order to compare their results to the present results.

Results are compared with the analytical solution that depends on the ratio between densities and viscosities, respectively denoted λ and ω:

$$\lambda = \frac{\rho_1}{\rho_2}, \quad \omega = \frac{\nu_1}{\nu_2} \tag{31}$$

We also define $\tilde{u}_1$ a characteristic velocity of the flow and Re_1 the corresponding Reynolds number:

$$\tilde{u}_1 = \frac{gL^2}{2\nu_1}, \quad Re_1 = \frac{\tilde{u}_1 L}{\nu_1} = \frac{gL^3}{2\nu_1^2} \tag{32}$$

Finally, we define the following dimensionless quantities:

$$y^+ = \frac{y}{L}, \quad u^+ = \frac{u_x}{\tilde{u}_1}, \tag{33}$$

where $u_x = \underline{u} \cdot \underline{e}_x$ with $\underline{u}$ the velocity of the fluid. With these notations, the analytical solution u^+_{th} reads:

$$u^+_{\mathrm{th}}(y^+) = \begin{cases} [1 - (y^+)^2] + \frac{1-\omega}{1+\lambda\omega}(y^+ - 1) & \text{if} \quad y^+ \in [0; 1] \\ \omega[1 - (y^+)^2] + \lambda\omega\frac{1-\omega}{1+\lambda\omega}(y^+ + 1) & \text{if} \quad y^+ \in [-1; 0] \end{cases} \tag{34}$$

Comparison of viscous terms ($Re_1 = 1.25$) *–*. Fig. 2 shows the longitudinal velocity profile obtained using Morris et al.'s (1997) viscous term (18), with $L/\delta r = 385$ SPH particles on the channel width. Numerical results are in excellent agreement with the analytical solution (34) represented by the solid line. Results obtained with Hu and Adams's (2006) viscous term (17) and Espanol and Revenga's (2003) one (20) also give such good qualitative results, thus demonstrating that the shear stresses at the wall and at the interface are correctly calculated by the present multi-phase model.

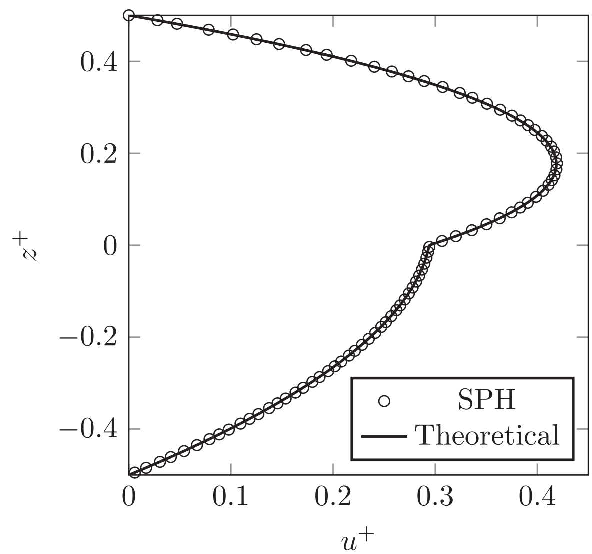

Fig. 2. Bi-fluid Poiseuille flow – horizontal velocity profile at the steady state for a Reynolds number of $Re_1 = 1.25$, with density and viscosity ratio of $\lambda = \omega = 4$ ($y^+ \leq 0 \Leftrightarrow$ fluid 1). Morris et al.'s (1997) viscous term is used and $L/\delta r = 384$. The black dots represent the SPH numerical result, while the solid line corresponds to the analytical solution (34).

Table 1
Bi-fluid Poiseuille flow – physical and numerical parameters for a Reynolds number of $Re_1 = 1.25$, with $\omega = \lambda = 4$.

			Fluid 1	Fluid 2
Phys.	g	[m · s^{-2}]	$4.0{\cdot}10^{-7}$	$4.0{\cdot}10^{-7}$
	ρ_0	[kg · m^{-3}]	4000	1000
	η	[Pa · s]	1.6	0.1
Num.	c_0	[m · s^{-1}]	0.02	0.02
	P_{bg}	[Pa]	0	0
	ξ	–	7	7
	Λ_{BP}	–	0.1	0.1

In order to validate further the multi-phase model, and to quantitatively compare the three viscous terms (18), (17) and (20), we performed a convergence study with the parameters summarized in Table 1. For each simulation, velocity field is initialized to zero, the numerical solution converges to a steady-state (see Fig. 3a) that is compared to the analytical solution (34) through the velocity instantaneous L_2 relative error E defined as follows:

$$E(t) = \sqrt{\frac{\left(\sum_{b\in\mathscr{F}} \underline{u}_b(t) - \underline{u}_{\mathrm{th}}(y_b, t)\right)^2}{\sum_{b\in\mathscr{F}} (\underline{u}_{\mathrm{th}}(y_b, t))^2}} \tag{35}$$

where $\underline{u}_b$ is the particle velocity, $\underline{u}_{\mathrm{th}}(y_b)$ the theoretical velocity at the position of particle b. The steady state is assumed to be achieved when the profile is fully developed. The characteristic time of the viscous effects propagation through the channel width L can be evaluated as $t_1 = L^2\rho_1/\eta_1$, from which we defined the dimensionless time $t^+ = t/t_1$. Fig. 3a shows the instantaneous error with respect to the dimensionless time. We can see that the steady state is achieved at $t^+ \approx 1$, and that the solution remains stable afterwards. In order to have an objective measure of the error, independent from the small oscillations of the instantaneous error, we define the time-averaged error $\overline{E}$, computed from (35) as:

$$\overline{E} = \frac{1}{N_t}\sum_{i=1}^{N_t} E(t_i) \tag{36}$$

where $(t_i)_{i\in\{1..N_t\}}$ corresponds to the N_t iterations in the time interval $t^+ \in [1.5, 3.5]$ (grey region in Fig. 3a).

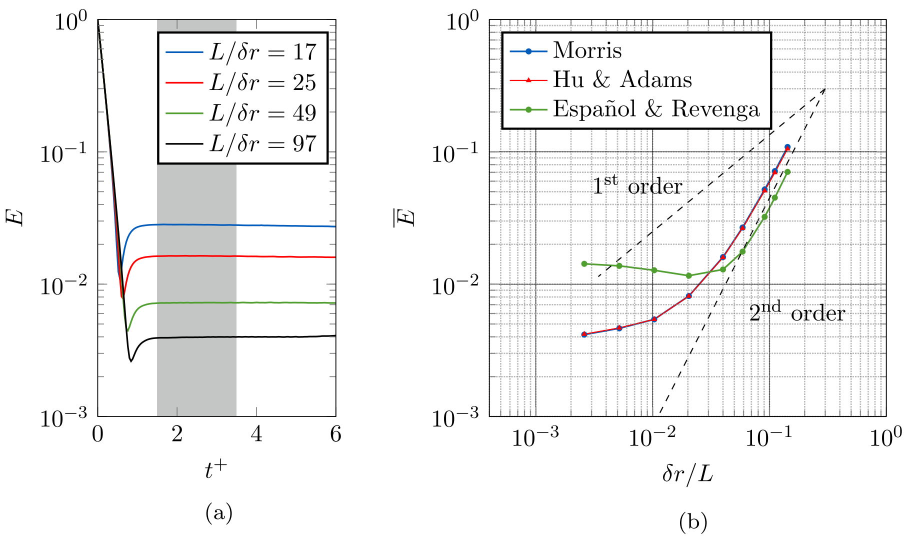

Fig. 3. Bi-fluid Poiseuille flow $Re_1 = 1.25$ – (a) evolution of the instantaneous L_2 relative error E with respect to the dimensionless time t^+. (b) Convergence graph of the present method comparing Morris et al.'s (1997) (18), Hu and Adams' (17) and Español and Revenga's (20) viscous terms, for nine values of $\delta r/L$.

Fig. 3b shows the time-average error $\overline{E}$ with respect to the dimensionless particle size $\delta r/L$ (*i.e.* the inverse of the number of particles along the channel height). We see that the convergence slope of the present method using the three viscous terms is approximately of order 2 as long as $\delta r/L \leq 5{\cdot}10^{-2}$. Afterwards, the convergence slope decreases for Morris et al.'s (1997) and Hu and Adams' operators, while the error stabilized for Espanol and Revenga's (2003) viscous term. These results are consistent with those obtained by Ferrand et al. (2013) for one fluid SPH formulation with USAW boundary conditions, thus validating the present multi-phase model. Note that, contrary to the results obtained by Ghaitanellis et al. (2015), Hu and Adams' operators is not less accurate than Morris's one. This is because there was an implementation error in Ghaitanellis et al.'s (2015) code.

3. Sediment constitutive model

3.1. Context

The main difficulty in modelling granular materials is that they cannot be classified as solids or liquids. Indeed they can behave like both of them under slightly different conditions. That is the reason why rheological approach is often adopted to model granular flows. In particular, soil is usually treated as a viscoplastic material. Thus it has a yield stress τ_y under which no deformation occurs. When the yield stress is exceeded, the material starts to flow. In such a model, the yield stress has to be compared to an invariant measure of the deviatoric stress tensor $\underline{\tau}$. In this work, the tensor second invariant is used according to Drucker–Prager yield criterion (Drucker and Prager, 1952):

$$\tau = \sqrt{\frac{1}{2}\sum_{i,j}\tau_{ij}\tau_{ij}} \tag{37}$$

Viscoplastic models include Herschel–Bulkley's and Casson's ones, but the most common and best-known model was proposed by Bingham (1917) and reads:

$$\begin{cases} \underline{\dot{\gamma}} = \underline{0} & \text{if} \quad \tau < \tau_y \\ \underline{\tau} = 2\left(\eta_\infty + \frac{\tau_y}{\dot{\gamma}}\right)\underline{\dot{\gamma}} & \text{if} \quad \tau > \tau_y \end{cases} \tag{38}$$

where η_∞ is the plastic viscosity, $\underline{\dot{\gamma}}$ the strain rate tensor and $\dot{\gamma}$ its second invariant defined as:

$$\dot{\gamma} = \sqrt{2\sum_{i,j}\dot{\gamma}_{ij}\dot{\gamma}_{ij}} \tag{39}$$

with $\dot{\gamma}_{ij}$ the components of the strain rate tensor defined as:

$$\underline{\dot{\gamma}} = \frac{1}{2}\left(\underline{\underline{\text{grad}}}\, \underline{u} + \left(\underline{\underline{\text{grad}}}\, \underline{u}\right)^T\right) \tag{40}$$

The ideal Bingham model (38) describes a material that behaves as a rigid body at low stresses and flows as a fluid when the yield stress is exceeded. Unfortunately, ideal Bingham's model, as well as Herschel–Bulkley's and Casson's ones, all involve discontinuous behaviour law that corresponds to the liquid-solid phase change. They are thus poorly adapted to numerical simulation because they require additional numerical procedures to track down yielded/unyielded regions. Indeed, it can be easily verified that the transitions between liquid and solid states are not included in the model. To do so, let us calculate τ applying (37) to the second equation in (38), which corresponds to the liquid state:

$$\tau = \eta_\infty \dot{\gamma} + \tau_y \tag{41}$$

One can see that when the strain rate vanishes, the deviatoric stress decreases and tends to τ_y but never goes below. To circumvent this issue, these models can be approximated assuming that the viscoplastic material is a liquid that exhibits infinitely high viscosity when the strain rate vanishes. Beverly and Tanner's (1992) bi-viscosity model and Papanastasiou's (1987) formula are two widely used examples of such regularized viscoplastic models. Thus the material is always liquid but mimics the ideal models behaviour for all rates of deformation. Numerically, this is achieved using shear thinning rheological law and limiting the viscosity to a huge but finite maximum value. This method is straightforward and has been widely applied to the simulation of sediment transport with Finite Volumes (Morichon et al., 2013), Moving Particle Semi-implicit method (Nabian and Farhadi, 2016) and smoothed particle hydrodynamics method (Capone et al., 2010).

Regarding sediment transport modelling, this approach is well adapted for highly dynamic scenarios but exhibits severe drawbacks regarding small strain-rates and deformations. Indeed, the maximum viscosity has to be large enough to guarantee that no significant motion occurs in unyielded regions. This is particularly important regarding erosion and scour development for which an accurate treatment of motionless regions is essential to correctly estimate the bed evolution. Thus, it is necessary to ensure that results do not depend on the chosen maximum viscosity. But in practice, the maximum value of viscosity is actually limited for explicit time integration schemes, because of computational cost.

As a consequence, in the present paper, a different approach is tested. The model of Ulrich (2013) is implemented in the framework of multi-phase formulation, and adapted to USAW boundary conditions. The soil is treated as a linear-elastic solid in unyielded regions, and as a shear thinning liquid in yielded regions. A continuous transition between the two states is ensured by a blending function driven by the strain rate magnitude. Besides, improvements of the yield stress and solid forces computation are proposed, as well as phase transition threshold based on the physical properties of the granular material.

3.2. Modelling assumptions

In this work, the granular material, being dry or saturated, is treated as a continuum. Under this assumption, the continuous medium represents a mixture of grains and air, or grains and water. Therefore, we need to define equivalent properties of the continuous material from the physical properties of the granular material components. The continuum properties should also depend on the grain concentration. However the continuum hypothesis presents a major difficulty within the framework of the SPH formulations presented in Section 2.2. Physically, the evolution of grain concentration results from differences of velocity between the solid phase and the liquid phase. When the mixture is assumed to be continuous, the concentration evolution must be taken into account through the transport of a concentration field. However, this is not straightforward in the framework of the present SPH model because this implies mass transfers between SPH particles. The particle masses being no longer constant, the SPH formulation presented in Section 2.2 is not valid anymore and a whole new SPH formulation would be necessary. Such a model is beyond the scope of this paper. Thus, although it has be shown that the grain concentration is an important parameter for the description of dense granular flows (Armanini et al., 2005; GDR MiDi, 2004), the grain concentration will be assumed to be constant and homogeneous within the mixture domain. Therefore, in practise, ϕ is simply the porosity of the granular material at initial time.

In addition, the fluid surrounding the grains will be assumed to be at rest with respect to the granular material skeleton. Indeed solving the fluid flow through the porous medium raises many practical issues, in particular at water-mixture interfaces. For example, let us consider an outgoing water flow from the mixture phase to the water phase. Then the mixture SPH particles should give some water mass up to the water SPH particles in order to ensure the mass conservation. Again, this would require variable SPH particle masses. Ulrich (2013) proposed a simplified model of partly saturated soil, solving the flow in the porous material and transporting a concentration field. However, with this model, the mass of water is not conserved. Consequently, in his work the water flow through the porous medium is also neglected and the interstitial water is assumed to move at the mixture velocity. This simplification implies several disadvantages. First, the seepage cannot be taken into account while it can have a significant effect when a stable structure of dry granular material becomes partially saturated (Liu and Li, 2015). Thus in the present work, we will only consider cases in which the material is initially totally dry (denoted by ^{dry}) or fully saturated (denoted by ^{sat}), and remains that way for ever. For the dry case, the mass of air in the pore is neglected, while for the saturated case, the equivalent density also depends on water density ρ_w:

$$\begin{cases} \rho_{eq}^{dry} = (1-\phi)\rho_g \\ \rho_{eq}^{sat} = (1-\phi)\rho_g + \phi\,\rho_w \end{cases} \tag{42}$$

Although a weakly-compressible formulation is used in this work, we aim at modelling incompressible materials. The equation of state (3) is a numerical way of closing the equations of motion system, but the compressible effects are not supposed to play any role in the behaviour of the granular material. Thus, we assume here that ρ_w and ρ_{eq}^{sat} are constant. Secondly, neglecting the flow through the porous material also implies that the pore water pressure field is unknown. Therefore, another approximation is necessary and the pore water pressure will be assumed to be hydrostatic.

Finally the granular medium will be considered incompressible. In soil mechanics, this hypothesis is only valid for fully saturated soils under fast loading (un-drained condition) (Craig, 1983; Davis and Selvadurai, 2005). Thus, as regards bed-load transport and scouring resulting from violent flows, this assumption is suitable. Although this hypothesis is not valid for problems involving dry granular media, we will see that the model is able to give satisfactory results in that case too.

3.3. Yield stress

There are many yield criteria that are adapted to the different kinds of viscoplastic materials. Although the Mohr-Coulomb criterion is widely used in geotechnical engineering, its implementation raises numerical issues because the corresponding yield surface is an irregular hexagonal pyramid (Jiang and Xie, 2011). Thus, the Drucker–Prager yield criterion is usually used as a smooth alternative. It is a pressure-dependent model that can be expressed in terms of internal friction angle ψ for non-cohesive soils as:

$$\tau_y = \frac{2\sqrt{3}\sin\psi}{3-\sin\psi}\,p_{eff} \tag{43}$$

where p_{eff} is the effective pressure, that is to say the normal stress that the grains actually exert on each other. Considering a situation where the stress configuration is close to the failure, almost no motion occurs within the material. Therefore, the isotropic stress (*i.e.* the total pressure p_{tot}) is solely due to pressure derived from the weight of the column of material and interstitial fluid above a specified level. Consequently, denoting z_i the soil-water interface vertical position, z_{fs} the free-surface vertical position and z the vertical position of the point of interest (see Fig. 4), the total pressure reads:

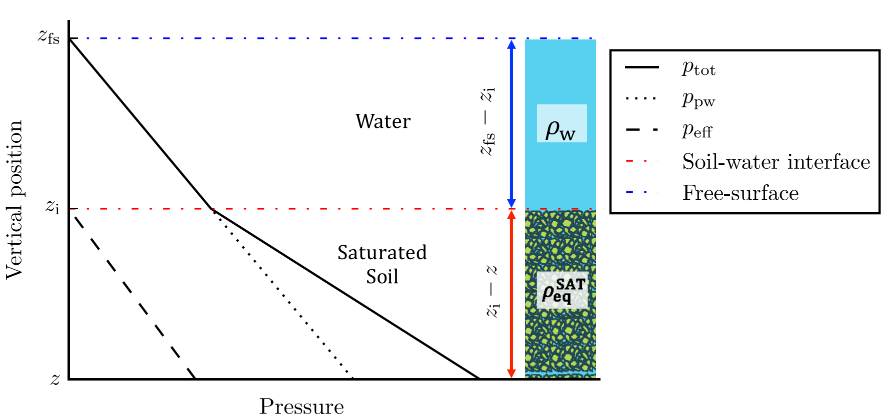

Fig. 4. Characteristic pressures in a submerged bed of soil saturated with water.

$$p_{\text{tot}} = \rho_{\text{w}} g (z_{\text{fs}} - z_{\text{i}}) + \rho_{\text{eq}}^{\text{sat}} g (z_{\text{i}} - z) \tag{44}$$

Moreover, Terzaghi's principle (Terzaghi, 1936) states that the effective pressure depends on the total pressure p_{tot} and the pore water pressure p_{pw} according to the following relation:

$$p_{\text{eff}} = p_{\text{tot}} - p_{\text{pw}} \tag{45}$$

Assuming no porous flow inside the soil, the pore water pressure reads:

$$p_{\text{pw}} = \rho_{\text{w}} g (z_{\text{fs}} - z) \tag{46}$$

Finally, effective pressure can be obtained using (45):

$$p_{\text{eff}} = g (z_{\text{i}} - z) \left(\rho_{\text{eq}}^{\text{sat}} - \rho_{\text{w}} \right) \tag{47}$$

Nevertheless, getting p_{eff} from (47) requires to detect soil-water interface beforehand. Such a detection has been tested by Manenti et al. (2011) but it seems to us that it can lead to inaccurate result when the interface is highly deformed. In order to get effective pressure without interface tracking, Fourtakas and Rogers (2016) proposed to use a modified state equation. However, we found that this method leads to a large overestimation of p_{eff} when the soil is submerged. Furthermore, with such an approach, the well known WCSPH pressure oscillations (Fatehi and Manzari, 2011) also impact the yield stress.

Therefore, we propose here another simple and reliable solution. Recall that ρ_{w} and $\rho_{\text{eq}}^{\text{sat}}$ are constant. In addition, z_{i} is the function that gives the mixture-water interface above the point of interest, so it does not depend on the vertical coordinate. As long as the water-mixture interface is not too much deformed, we can besides assume that the interface varies linearly in the horizontal directions in the vicinity of the point of interest (this is the strongest assumption done in this reasoning). Thus, taking the Laplacian of (47) yields:

$$\Delta p_{\text{eff}} = 0 \tag{48}$$

taking the Laplacian of (47) with suitable boundary conditions we get the following system:

$$\begin{cases} \Delta p_{\text{eff}}(\underline{r}) = 0 & \text{for} \quad \underline{r} \in \Omega \\ p_{\text{eff}}(\underline{r}) = 0 & \text{for} \quad \underline{r} \in \mathscr{I} \\ \frac{\partial p_{\text{eff}}}{\partial \underline{n}}(\underline{r}) = \underline{g} \cdot \underline{n} & \text{for} \quad \underline{r} \in \mathscr{S} \end{cases} \tag{49}$$

with Ω the simulation domain, $\mathscr{S}$ the solid boundaries, $\underline{n}$ the inward boundary normal and $\mathscr{I}$ the boundary corresponding to soil free-surface and soil-water interface (see Fig. 4).

In the present model p_{eff} is only used to compute the yield stress. It cannot be used in the momentum conservation equation, because (49) is valid when the total pressure is approximately lithostatic, which is not true in yielded regions. Thus, the total pressure calculated with the state Eq. (3) is used to compute the pressure gradient in the momentum conservation equation. This amounts to assume that the pore water pressure gradient can be neglected compared to the effective pressure gradient:

$$\underline{\nabla} p_{\text{eff}} = \underline{\nabla} p_{\text{tot}} - \underline{\nabla} p_{\text{pw}} \approx \underline{\nabla} p_{\text{tot}} \tag{50}$$

Note that it would be possible to eliminate this approximation observing that (49) can easily be adapted to compute the hydrostatic pore water pressure. The pressure gradient in the momentum conservation equation could thus be computed from $p_{\text{tot}} - p_{\text{pw}}$. This was not tested in this work.

3.4. Shear stresses

3.4.1. Elastic solid state

In the present model, the solid state of the granular material is evaluated in line with linear elastic theory. The shear stress tensor in the solid regions reads:

$$\underline{\underline{\tau}}_{\text{g}}^{\text{s}} = 2 \mathscr{G} \underline{\underline{\gamma}} \tag{51}$$

can be obtained from the following rate of stress equation:

$$\dot{\underline{\underline{\tau}}}_{\text{g}}^{\text{s}} = 2 \mathscr{G} \dot{\underline{\underline{\gamma}}} \tag{52}$$

where the dot · denotes the time derivative, the subscript $_{\text{g}}$ refers to the granular material, and the superscripts $^{\text{s}}$ refers to the elastic-solid model, *i.e.* the solid state and $\mathscr{G}$ is the shear modulus defined by:

$$\mathscr{G} = \frac{E}{2(1+\nu)} \tag{53}$$

with E the Young's modulus and ν the Poisson's coefficient. Note that, similarly to many authors in SPH literature (Gray et al., 2001; Randles and Libersky, 1996; Ulrich, 2013), Hooke's law will be used to model the deviatoric part of the stress tensor, while an equation of state will be used to compute the pressure according to the weakly-compressible approach (see Section 2). As mentioned in Section 2.1, here the compressibility is only a numerical trick to mimic the incompressible behaviour. This is consistent with the assumption of an incompressible linear-elastic material (see Section 3.2). Note that such a state equation has already been successfully used to compute the pressure in a linear elastic material in the SPH literature, *e.g.* in Gray et al. (2001) and Ulrich (2013).

In (52) we differentiated the solid state behaviour law (51) for the following reason. It can be shown that the rate of stress (52) is not objective, *i.e.* it is not independent on the frame of reference. In particular, it is not invariant with respect to rigid-body rotation. Though, the material constitutive equations should be frame indifferent since the mechanical response of a material must not depend on the observer. Thus the dynamic quantities, such as the Cauchy stress tensor, are frame invariant (Dienes, 1979; Gurtin, 1981). To circumvent this issue, objective material time derivatives must be used. The most commonly used in computational mechanics is the Jaumann time derivative, denoted with a triangle $^{\Delta}$ and defined by:

$$\overset{\Delta}{\underline{\underline{\sigma}}} = \dot{\underline{\underline{\sigma}}} + \underline{\underline{\sigma}}\, \dot{\underline{\underline{\omega}}} - \dot{\underline{\underline{\omega}}}\, \underline{\underline{\sigma}} \tag{54}$$

where $\dot{\underline{\underline{\omega}}}$ is the rotation rate tensor that is directly related to the rate of rigid-body rotation within a material:

$$\dot{\underline{\underline{\omega}}} = \frac{1}{2} \left(\underline{\underline{\text{grad}}}\ \underline{u} - \left(\underline{\underline{\text{grad}}}\ \underline{u} \right)^{T} \right) \tag{55}$$

Note that, when no rigid-body rotation occurs within the material, the Jaumann material derivative (54) is tantamount to the standard material time derivative. Applying (54) to the deviatoric stress tensor $\underline{\underline{\tau}}$ yields:

$$\overset{\Delta}{\underline{\underline{\tau}}}{}_{\text{g}}^{\text{s}} = 2 \mathscr{G} \dot{\underline{\underline{\gamma}}} + \dot{\underline{\underline{J}}} \tag{56}$$

denoting $\dot{\underline{\underline{J}}}$ the Jaumann rate tensor defined by:

$$\dot{\underline{\underline{J}}} = \underline{\underline{\tau}}_{\text{g}}^{\text{s}}\, \dot{\underline{\underline{\omega}}} - \dot{\underline{\underline{\omega}}}\, \underline{\underline{\tau}}_{\text{g}}^{\text{s}} \tag{57}$$

The shear stress tensor finally reads:

$$\underline{\underline{\tau}}_{\text{g}}^{\text{s}} = 2 \mathscr{G} \underline{\underline{\gamma}} + \underline{\underline{J}} \tag{58}$$

which is more relevant than the initial attempt (51).

3.4.2. Viscoplastic fluid state

In yielded regions, the material is modelled as a shear thinning liquid. Thus the viscous fluid behaviour law (Stokes, 1851) is used to model the shear stress:

$$\underline{\underline{\tau}}_{\text{g}}^{\text{l}} = 2 \eta_{\text{eff}} \dot{\underline{\underline{\gamma}}} \tag{59}$$

where the superscript l refers to the viscoplastic model, *i.e.* the liquid state. In Ulrich's (2013) original model, the effective viscosity η_{eff} was calculated according to a rheological law similar to the $\mu(I)$ rheology (Jop et al., 2006). Here, we propose a different law in order to be consistent with the Drucker–Prager yield criterion. The effective viscosity is thus defined by:

$$\eta_{\text{eff}} = \frac{\tau_{\text{y}}}{\dot{\gamma}} \tag{60}$$

where τ_{y} is the yield stress calculated according to (43). Therefore, similarly to the $\mu(I)$ rheology (Jop et al., 2006; Revil-Baudard and Chauchat, 2013), the effective viscosity is related to the friction within the granular material. However, here the friction coefficient is constant in time since the internal friction ψ is a constant parameter in our model. It should be noted that (60) ensures the continuity of the shear stress magnitude in the liquid-solid transition since we have:

$$\tau_{\text{g}}^{\text{l}} = \sqrt{\frac{1}{2}\underline{\underline{\tau}}_{\text{g}}^{\text{l}} : \underline{\underline{\tau}}_{\text{g}}^{\text{l}}} = \eta_{\text{eff}}\,\dot{\gamma} = \tau_{\text{y}} \tag{61}$$

3.4.3. Solid-fluid transition

Similarly to the ideal viscoplastic models (Mitsoulis, 2007) (Bingham, Herschel–Bulkley, Casson), the elastic-viscoplastic model (Ulrich, 2013) involves a discontinuous behaviour law. Thus, the liquid-solid phase transition cannot solely be based on the yield stress. Consider the material is in the liquid state, then the shear stress magnitude is always equal to τ_{y}, as shows in (61). Thus the material cannot become solid. As a consequence, similarly to regularized viscoplastic models (Papanastasiou, 1987; Tanner and Milthorpe, 1983), the yield criterion is not based on a dynamic condition anymore ($\tau > \tau_{\text{y}}$), but on a kinematic condition ($\dot{\gamma} > \dot{\gamma}_{\text{y}}$). Thus we need to build a new threshold expressed in terms of strain rate that still depends on the yield stress. To do so we write:

$$\dot{\gamma}_{\text{y}} = \frac{\tau_{\text{y}}}{\eta_{\text{y}}} \tag{62}$$

where η_{y} is a dynamic viscosity referred to as yield viscosity. Thus, (62) points out that the solid-liquid transition now depends on an additional parameter η_{y}. Moreover, the yield viscosity not only determines the threshold strain rate but also the maximum possible viscosity for the material in liquid state.

In Ulrich (2013), the value of η_{y} is chosen in a trial and error approach, increasing incrementally its value until obtaining satisfying result. Here we propose to find a value based on the physical properties of the material. To do so, we first consider the Deborah number *De* which characterizes the fluidity of a material under the flow conditions of a specific experiment:

$$De = \frac{t_R}{t_C} \tag{63}$$

where t_C is the characteristic time scale of the experiment, and t_R the stress relaxation time. For a granular material, assuming spherical grains, Sun and Wang (2013) proposed to use the characteristic time of Rayleigh waves propagating along a grain surface:

$$t_R^G = \frac{\pi d_{\text{g}}}{0.162\nu + 0.887}\sqrt{\frac{\rho_{\text{g}}}{\mathscr{G}}} \tag{64}$$

with d_{g} the grain diameter. On the other hand, during the transition, the soil can be seen as a viscoelastic material. In that case, the Deborah number is calculated with a relaxation time defined as:

$$t_R^{VE} = \frac{\eta_{\text{y}}}{\mathscr{G}} \tag{65}$$

For a given material, the two relaxation time definitions should match. Therefore, combining (64) and (65) gives a definition of the yield viscosity:

$$\eta_{\text{y}} = \frac{\pi d_{\text{g}}\sqrt{\mathscr{G}\rho_{\text{g}}}}{0.163\nu + 0.877} \tag{66}$$

We can now use the blending function proposed by Ulrich (2013):

$$\zeta(\dot{\gamma}) = \begin{cases} (q+1)\left(\frac{\dot{\gamma}}{\dot{\gamma}_{\text{y}}}\right)^q - q\left(\frac{\dot{\gamma}}{\dot{\gamma}_{\text{y}}}\right)^{q+1} & \text{if } \dot{\gamma} < \dot{\gamma}_{\text{y}} \\ 1 & \text{if } \dot{\gamma} \geq \dot{\gamma}_{\text{y}} \end{cases} \tag{67}$$

with q a positive integer. The solid-liquid transition sharpness can be adjusted varying q. Similarly to Ulrich (2013), we use $q = 2$. Note that the blending function has the same role as Papanastasiou's (1987) regularization. It smooths the phase transition, which is more appropriate than a sharp interface for numerical modelling. The liquid-solid deviatoric stress tensor is thus given by:

$$\underline{\underline{\tau}} = \zeta\,\underline{\underline{\tau}}_{\text{g}}^{\text{l}} + (1-\zeta)\,\underline{\underline{\tau}}_{\text{g}}^{\text{s}} \tag{68}$$

It is important to note that the solid stresses in the soil should vanish when the liquid state is reached. After flowing, the grains have no memory of their initial position. Thus, the strain tensor must be re-initialized when $\zeta \to 1$.

Furthermore, the phase transition is based on a condition on the strain rate magnitude $\dot{\gamma}$ while the elastic solid stresses do not depend on the strain rate. Thus, there can be situations where the liquid state is not fully reached ($\dot{\gamma} < \dot{\gamma}_{\text{y}}$), but where the solid stress magnitude exceeds the yield stress ($\tau > \tau_{\text{y}}$). This happens when the failure is followed by very slow deformations. Thus the strain rate $\dot{\gamma}$ remains smaller than $\dot{\gamma}_{\text{y}}$ but the strain keeps increasing. Consequently, the solid stress magnitude also keeps increasing and ends up exceeding the yield stress. However, the material is supposed to yield when the stress magnitude reaches the yield stress, so this never happens physically. To avoid unphysical overestimation of the solid stresses, we need to rescale the solid stress tensor so that the solid stress magnitude never exceeds the yield stress:

$$\begin{cases} \underline{\underline{\tau}}_{\text{g}}^{\text{ss}} = \underline{\underline{\tau}}_{\text{g}}^{\text{s}} & \text{if } \tau_{\text{g}}^{\text{s}} \leq \tau_{\text{y}} \\ \underline{\underline{\tau}}_{\text{g}}^{\text{ss}} = \frac{\tau_{\text{y}}}{\tau}\underline{\underline{\tau}}_{\text{g}}^{\text{s}} & \text{if } \tau_{\text{g}}^{\text{s}} > \tau_{\text{y}} \end{cases} \tag{69}$$

where $\underline{\underline{\tau}}_{\text{g}}^{\text{ss}}$ denotes the rescaled deviatoric stress tensor. The final form of the liquid-solid deviatoric stress tensor is thus:

$$\underline{\underline{\tau}} = \zeta\,\underline{\underline{\tau}}_{\text{g}}^{\text{l}} + (1-\zeta)\,\underline{\underline{\tau}}_{\text{g}}^{\text{ss}} \tag{70}$$

Finally, in this model the mechanical behaviour of the soil only depends on elastic (*E*, ν) and granular (d_{g}, ρ_{g}, ϕ, ψ) properties of the material, especially through (43) and (66) that drive the liquid-solid transition.

4. SPH implementation

In SPH the continuum is sampled in a set of interpolation points $\mathscr{F}$ having a constant mass m_a and referred to as *particles*. An SPH particle used to discretize a granular material continuum represents a small volume that contains a mixture of solid grains and surrounding fluid. The equivalent density of the continuum is calculated from (42). The present granular material model is implemented within the scope of the multi-phase SPH formulation presented in Section 2.2. First, let us define (or recall) the following sets of SPH particles:

- $\mathscr{V}$ is the set of vertex particles (see USAW framework in Section 2.2.2)
- $\mathscr{S}$ is the set of boundary elements (see USAW framework in Section 2.2.2)
- $\mathscr{M}$ is the set of free particles of granular material (mixture of grains and surrounding fluid)
- $\mathscr{W}$ is the set of free particles of water
- $\mathscr{F}$ is the set of free particles of matter: $\mathscr{F} = \mathscr{M} \cup \mathscr{W}$
- $\mathscr{I}$ is the set of free particles of matter at the water-mixture interface:

$\mathscr{I} \subset \mathscr{F}$

4.1. Effective pressure

To compute the effective pressure p_{eff}, we need to solve the Laplace Eq. (49). Thus we use Morris' Laplacian (18) to write the corresponding discrete equation:

$$\forall\, a \in (\mathscr{M} \setminus \mathscr{I}) \quad L_a^U\{1, p_{\text{eff},b}\} = 0 \tag{71}$$

where $(\mathscr{M} \setminus \mathscr{I})$ refers to the set of SPH particles of mixture that are not located at the interface. At the wall, the Neumann boundary conditions (49) must be satisfied. Applying the technique used by Ferrand et al. (2013) to compute the dynamic pressure at the wall, we use a Shepard-like interpolation of the effective pressure field:

$$\forall\, a \in (\mathscr{V} \cup \mathscr{S}),\ p_{\text{eff},a} = \frac{1}{\sum_{b\in\mathscr{F}_S} V_b w_{ab}} \left(\sum_{b\in\mathscr{F}_S} V_b (p_{\text{eff},b} + \Delta\rho_{\text{eq}}\, \underline{g}\cdot(\underline{r}_a - \underline{r}_b)) w_{ab} \right) \tag{72}$$

with $(\mathscr{V} \cup \mathscr{S})$ the set of boundary elements and vertex particles. Besides, a particular attention must be paid to the Dirichlet condition imposed to soil particles located at the soil-water interface (second line of (49)). Indeed, the pressure due to their own weight must be taken into account, otherwise the yield stress (43) is always zero at interface particles:

$$\forall\, a \in (\mathscr{M} \cap \mathscr{I}) \quad p_{\text{eff},a} = g\delta r\, \Delta\rho_{\text{eq}} \tag{73}$$

with $(\mathscr{M} \cap \mathscr{I})$ the set of SPH particles of mixture located at the interface, and δr the initial particle spacing (*i.e.* the particle size). Finally, (71) is solved using a Jacobi solver initialized with the previous time step solution, thus ensuring low computational cost.

4.2. Shear forces

4.2.1. Elastic solid force density

To compute elastic forces, we first need to compute the strain rate tensor $\underline{\underline{\dot{\gamma}}}$ and the rotation rate tensor $\underline{\underline{\dot{\omega}}}$:

$$\underline{\underline{\dot{\gamma}}}_a = \frac{1}{2}(\underline{\underline{G}}_a\{\underline{u}_b\} + (\underline{\underline{G}}_a\{\underline{u}_b\})^T) - \frac{1}{3}\text{Tr}(\underline{\underline{G}}_a\{\underline{u}_b\})\,\underline{\underline{I}} \tag{74}$$

$$\underline{\underline{\dot{\omega}}}_a = \frac{1}{2}(\underline{\underline{G}}_a\{\underline{u}_b\} + (\underline{\underline{G}}_a\{\underline{u}_b\})^T) \tag{75}$$

where the multi-phase anti-symmetric gradient based on (16) is used here:

$$\underline{\underline{G}}_a\{\underline{u}_b\} = -\frac{\overline{V}_a}{\Gamma_a}\sum_{b\in(\mathscr{F}\cup\mathscr{V})} \theta_b(\underline{u}_a - \underline{u}_b) \otimes \underline{\nabla} w_{ab} + \frac{\overline{V}_a}{\Gamma_a}\sum_{s\in\mathscr{S}} \frac{1}{\overline{V}_s}(\underline{u}_a - \underline{u}_s) \otimes \underline{\nabla}\Gamma_{as} \tag{76}$$

with $\otimes$ denoting the tensor product. Note that the anti-symmetric gradient (76) is used instead of (14) for more accuracy. It should also be reminded that the trace of the velocity gradient should be zero for an incompressible fluid. But within the scope of WCSPH, we have to remove the trace from the strain rate tensor in (74), to ensure that the deviatoric stress tensor $\underline{\underline{\tau}}$ is really traceless. The shear stress tensor then follows from an explicit time-integration of the Jaumann rate of shear stress (56):

$$\underline{\underline{\tau}}_{\text{g}}^{\,\text{s}} = \overline{2\mathscr{G}\,\underline{\underline{\dot{\gamma}}} + \underline{\underline{\tau}}\,\underline{\underline{\dot{\omega}}} - \underline{\underline{\dot{\omega}}}\,\underline{\underline{\tau}}} \tag{77}$$

where the overbar indicates explicit time integration. Then, the elastic solid force density could be simply computed from the divergence of the deviatoric stress tensor:

$$\underline{\text{div}}\,\underline{\underline{\tau}}_{\text{g}}^{\,\text{s}} = \underline{\text{div}}(2\mathscr{G}\,\underline{\underline{\gamma}}) + \underline{\text{div}}\,\underline{\underline{J}} \tag{78}$$

Numerically, this implies two iterative use of first order SPH derivative operators. Indeed, (77) requires computing the strain rate tensor $\underline{\underline{\dot{\gamma}}}$ using the SPH gradient of velocity. Then, (78) requires to take the SPH divergence of the shear stress tensor (Gray et al., 2001; Ulrich, 2013). Thus, focusing on the first term in (78) yields:

$$(\underline{\text{div}}(2\mathscr{G}\,\underline{\underline{\gamma}}))_a \approx \mathscr{G}\,\underline{D}_a\{\overline{(\underline{\underline{G}}_b\{\underline{u}_c\} + (\underline{\underline{G}}_b\{\underline{u}_c\})^T)}\} \tag{79}$$

But due to the collocated nature of SPH, applying iteratively first order derivative operators to get higher order derivatives leads to significant inaccuracy (see Section 2.2.3). This can be avoided observing that the shear strain tensor γ can be calculated from the displacement $\underline{X}$ instead of the time-integration of the strain rate tensor:

$$\underline{\underline{\gamma}} = \frac{1}{2}\left(\underline{\underline{\text{grad}}}\,\underline{X} + \left(\underline{\underline{\text{grad}}}\,\underline{X}\right)^T\right) \tag{80}$$

This definition of shear strain tensor (80) is particularly adapted to SPH whose Lagrangian characteristic gives a direct access to particle displacement:

$$\underline{X}(t) = \underline{r}(t) - \underline{r}_0 \tag{81}$$

with $\underline{r}$ the particle position and $\underline{r}_0$ the initial particle position. Note that the initial particle position is reinitialised every time the material passes from liquid to solid state: after flowing, a grain has no memory of its initial position. Then the shear strain divergence can be obtained using Espanol and Revenga's (2003) second order derivative operator $\underline{L}^E$ defined in (20). Contrary to (9) and (18), Espanol and Revenga's (2003) formula (20) is not an approximation of the Laplacian (see (21)).

Now, applying (20) to shear modulus $\mathscr{G}$ and displacement $\underline{X}$ we get:

$$(\underline{\text{div}}(2\mathscr{G}\,\underline{\underline{\gamma}}))_a \approx \mathscr{G}\,\underline{L}_a^E\{1, \underline{X}_b\} \tag{82}$$

A similar treatment of the Jaumann term does not seem to be possible, so the shear force density f finally reads:

$$\left(\underline{f}_{\text{g}}^{\,\text{s}}\right)_a = \mathscr{G}\,\underline{L}_a^E\{1, \underline{X}_b\} + \underline{D}_a\{\underline{\underline{J}}_b\} \approx \left(\underline{\text{div}}\,\underline{\underline{\tau}}_{\text{g}}^{\,1}\right)_a \tag{83}$$

where the tensor $\underline{\underline{J}}$ follows from an explicit time-integration of the Jaumann rate tensor (57):

$$\underline{\underline{J}}^{(n+1)} = \underline{\underline{J}}^{(n)} + \Delta t\left(\underline{\underline{\tau}}_{\text{g}}^{\,\text{s}}\,\underline{\underline{\dot{\omega}}} - \underline{\underline{\dot{\omega}}}\,\underline{\underline{\tau}}_{\text{g}}^{\,\text{s}}\right)^{(n)} \tag{84}$$

The force density $\underline{f}_{\text{g}}^{\,\text{s}}$ is rescaled afterwards according to (69) (here we omit the SPH particle labels for the sake of clarity):

$$\begin{cases} \underline{f}_{\text{g}}^{\,\text{ss}} = \underline{f}_{\text{g}}^{\,\text{s}} & \text{if} \quad \tau_{\text{g}}^{\text{s}} \le \tau_{\text{y}} \\ \underline{f}_{\text{g}}^{\,\text{ss}} = \frac{\tau_{\text{y}}}{\tau_{\text{g}}^{\text{s}}}\underline{f}_{\text{g}}^{\,\text{s}} & \text{if} \quad \tau_{\text{g}}^{\text{s}} > \tau_{\text{y}} \end{cases} \tag{85}$$

Note that the latter rescaling is not tantamount to (69). Indeed, here we rescale forces instead of stresses, which means that the following approximation is done:

$$\text{div}\left(\frac{\tau_{\text{y}}}{\tau_{\text{g}}^{\text{s}}}\underline{\underline{\tau}}_{\text{g}}^{\,\text{s}}\right) \approx \frac{\tau_{\text{y}}}{\tau_{\text{g}}^{\text{s}}}\text{div}\left(\underline{\underline{\tau}}_{\text{g}}^{\,\text{s}}\right) \tag{86}$$

This is mandatory when using (82) instead of (79) because the stress tensor $\underline{\underline{\tau}}_{\text{g}}^{\,\text{s}}$ is not directly used to compute the forces. Although it will be seen that using (82) improves the computation of elastic forces (see Section 4.3), it would be necessary to evaluate the effect of the latter approximation on the model accuracy to ascertain whether (82) is globally better. Unfortunately, such an investigation has not been carried out in the scope of this work.

4.2.2. Viscous force density

When the material is in the viscoplastic liquid state, the force density is given by:

$$\left(\underline{\mathrm{div}}\ \underline{\underline{\tau}}_{\mathrm{g}}^{1}\right)_{a} = (\underline{\mathrm{div}}(2\eta_{\mathrm{eff}}\ \underline{\underline{\dot{\gamma}}}))_{a} \tag{87}$$

Thus, it is computed applying (20) to the effective viscosity η_{eff} and the velocity $\underline{u}$:

$$\left(\underline{f}_{\mathrm{g}}^{1}\right)_{a} = \underline{L}_{a}^{E}\{\eta_{\mathrm{eff},b},\ \underline{u}_{b}\} \approx (\underline{\mathrm{div}}(2\eta_{\mathrm{eff}}\ \underline{\underline{\dot{\gamma}}}))_{a} \tag{88}$$

Here again, Espanol and Revenga's (2003) operator is used instead of Morris' viscous term. This is due to the fact that the transposed velocity gradient in (40) cannot be neglected when modelling non-Newtonian fluids. The effective viscosity is computed according to (60) that depends on the yield stress (43) and the mean scalar rate of strain $\dot{\gamma}$ (39).

4.2.3. Blending of shear forces

The continuous transition between solid and liquid state is ensured by the blending function ζ (67) defined in Section 3.4.3 (here again we omit the SPH particle labels):

$$\underline{f}_{\mathrm{g}} = \zeta\underline{f}_{\mathrm{g}}^{1} + (1-\zeta)\underline{f}_{\mathrm{g}}^{s} \tag{89}$$

where $\underline{f}_{\mathrm{g}}^{s}$ and $\underline{f}_{\mathrm{g}}^{1}$ are computed respectively through (83) and (87). Note that, here the blending applies to the forces instead of the stresses, even though (89) is not tantamount to (67) since ζ in not constant in space. However, the blending being a numerical trick to ensure a continuous transition, (89) can be used too. A simple way of getting (67) back would be to use:

$$\left(\underline{f}_{\mathrm{g}}\right)_{a} = \underline{L}_{a}^{E}\{\zeta_{b}\eta_{\mathrm{eff},b},\ \underline{u}_{b}\} + \mathscr{G}\,\underline{L}_{a}^{E}\{(1-\zeta_{b}),\ \underline{X}_{b}\} + \underline{D}_{a}\{(1-\zeta_{b})\underline{\underline{J}}_{b}\} \tag{90}$$

However, the differences between (89) and (90) were not investigated in this work.

4.2.4. Boundary conditions

4.2.4.1. Conditions at water–granular-material interface. In the frame of the multi-phase formulation presented in Section 2.2, the density and the pressure fields do not require any special treatment for the interaction of SPH particles of water $\mathscr{W}$ and of granular material $\mathscr{M}$. The density is computed according to (13) from all the neighbouring particles, irrespective of the phase ($\mathscr{W}$ or $\mathscr{M}$) they belong to. Besides that, both phases have a pressure field, so the pressure gradient is simply computed with the multi-phase gradient (14).

In regard to the shear forces, the two phases only interact though viscous effects for which Espanol and Revenga's (2003) viscous term (20) is used in the two phases. In water, the granular material neighbours are seen as liquid particles having a viscosity η_{eff}. Thus the force density exerted by a particle b of granular material $\mathscr{M}$ on a particle a of water $\mathscr{W}$ reads:

$$\forall\ a\in\mathscr{W},\ b\in\mathscr{M},\ (\underline{f}_{\mathrm{w}})_{b\to a} = \theta_{b}\overline{V}_{b}\frac{\eta_{\mathrm{w},a}\,\eta_{\mathrm{eff},b}}{\eta_{\mathrm{w},a}+\eta_{\mathrm{eff},b}}[(d+2)((\underline{u}_{a}-\underline{u}_{b})\cdot \underline{e}_{ab})\underline{e}_{ab} + (\underline{u}_{a}-\underline{u}_{b})]\frac{\underline{\nabla}w_{ab}\cdot\underline{r}_{ab}}{r_{ab}^{2}} \tag{91}$$

where the subscript $_{\mathrm{w}}$ refers to the water and d is the space dimension. On the other hand, the viscous force density exerted by the water particle on the granular material contribute to the viscoplastic term in (89) through:

$$\forall\ a\in\mathscr{M},\ b\in\mathscr{W},\ \left(\underline{f}_{\mathrm{g}}^{1}\right)_{a\to b} = \zeta\left[\theta_{b}\overline{V}_{b}\frac{\eta_{\mathrm{w},a}\,\eta_{\mathrm{eff},b}}{\eta_{\mathrm{w},a}+\eta_{\mathrm{eff},b}}[(d+2)((\underline{u}_{a}-\underline{u}_{b})\cdot \underline{e}_{ab})(-\underline{e}_{ab}) - (\underline{u}_{a}-\underline{u}_{b})]\frac{\underline{\nabla}w_{ab}\cdot\underline{r}_{ab}}{r_{ab}^{2}}\right] \tag{92}$$

It should be noted that, since there is no contribution of the water neighbour the elastic-solid term in (89), the action-reaction principle is not respected when $\zeta < 1$. Indeed, from the two latter equations, we can see that we only have:

$$\forall\ a\in\mathscr{M},\ b\in\mathscr{W}\qquad (\underline{f}_{\mathrm{w}})_{b\to a} = -\frac{1}{\zeta}\left(\underline{f}_{\mathrm{g}}^{1}\right)_{a\to b} \tag{93}$$

Nevertheless, the strain rate of granular material particles close to the interface is usually high, due to the water particles velocity that contribute (74). Therefore, for particles in $\mathscr{M}$ having neighbours in $\mathscr{W}$, we usually have $\zeta\to 1$ so the action-reaction principle is respected most of the time. Despite that, (93) remains a weakness of the present model, and further investigation should be carried out to improve this aspect.

As a consequence of (91) and (92), the suspended sediment and water only exert viscous forces on each other. When the sediment re-settles, viscoplastic and elastic forces exert between sediment SPH particles.

As regards erosion, the water-sediment interaction could be improved taking account for turbulence. In the present model, the shear stress at the water-sediment interface depends on the shear rate, the dynamic viscosity of water, and the local viscoplastic viscosity of the sediment. Physically, erosion results from the viscous and turbulent shear stresses exerted by water on the solid grains of sediment. Thus, it may be possible to use and SPH form of the standard $k-\epsilon$ model in the water phase (Ferrand et al., 2013), treating the water-sediment interface as a solid wall. The stress exerted on the sediment SPH particles would then be deduced from the action-reaction principle. That way, the bottom shear stress would not depend on the rheological law used within the granular material. This approach has not been tested here.

Conditions at wall–granular-material interface. Wall boundary conditions must enforced on both the viscoplastic term (88) and the elastic solid term (83). With regard to the former, the method proposed by Ferrand et al. (2013) for the viscous wall condition is simply applied. As regards the elastic solid term, the displacement of vertex $\mathscr{V}$ and boundary elements $\mathscr{S}$ et imposed to zero. We recall Espanol and Revenga's (2003) boundary term:

$$\underline{L}_{a}^{E,\mathrm{bound}}\{1,\ \underline{X}_{b}\} = -\frac{1}{\Gamma_{a}}\sum_{s\in\mathscr{S}}(\underline{\underline{\nabla}}\underline{X}_{a} + \underline{\underline{\nabla}}\underline{X}_{s})\cdot\underline{\nabla}\Gamma_{as} \tag{94}$$

Using a Taylor expansion, we can write:

$$\begin{aligned}(\underline{\underline{\nabla}}\underline{X})_{a}\cdot\underline{e}_{as} &\approx \frac{\underline{X}_{a}-\underline{X}_{s}}{r_{as}}\\ (\underline{\underline{\nabla}}\underline{X})_{s}\cdot\underline{e}_{as} &\approx \frac{\underline{X}_{a}-\underline{X}_{s}}{r_{as}}\end{aligned} \tag{95}$$

Thus, imposing $\underline{X}_{s} = \underline{0}$, Espanol and Revenga's (2003) boundary term (94) can be approximated as:

$$\underline{L}_{a}^{E,\mathrm{bound}}\{1,\ \underline{X}_{b}\} \approx -\frac{2}{\Gamma_{a}}\sum_{s\in\mathscr{S}}\underline{X}_{a}\frac{\underline{r}_{as}\cdot\underline{\nabla}\Gamma_{as}}{r_{as}^{2}} \tag{96}$$

The latter approximation is not exact because $\underline{\nabla}\Gamma_{as}$ is a vector oriented along the inward wall normal vector, not along $\underline{r}_{as}$.

Then, we assume a locally uniform distribution of $\underline{\underline{J}}$ at the walls (recall J is the time anti-derivative of the Jaumann tensor, as defined in Section 3.4.1). The corresponding boundary term reads:

$$\underline{D}_{a}^{\mathrm{bound}}\{\underline{\underline{J}}_{b}\} = \frac{\overline{V}_{a}}{\Gamma_{a}}\sum_{s\in\mathscr{S}}\frac{1}{\overline{V}_{s}}(\underline{\underline{J}}_{a}-\underline{\underline{J}}_{s})\cdot\underline{\nabla}\Gamma_{as} \tag{97}$$

The present assumption yields $\underline{\underline{J}}_{s} = \underline{\underline{J}}_{a}$, hence:

$$\underline{D}_{a}^{\mathrm{bound}}\{\underline{\underline{J}}_{b}\} = \underline{0} \tag{98}$$

Similarly, we assume $\underline{\underline{J}}_{b} = \underline{\underline{J}}_{a}$ for vertex particles ($b\in\mathscr{V}$), thus the contribution of vertex particles to the volumic term also vanishes. The divergence of Jaumann rate finally simplifies to:

$$\underline{D}_a\{\underline{J}_b\} = -\frac{\overline{V}_a}{\Gamma_a}\sum_{b\in\mathscr{F}}\theta_b(\underline{J}_a - \underline{J}_b)\cdot\underline{\nabla}w_{ab} \tag{99}$$

where the boundary terms vanished and the sum now only extends to free particles $\mathscr{F}$.

4.3. Validation of elastic forces computation

This test case is a simple numerical experiment that aims at testing the improvement resulting from the computation of solid forces from the displacement field, *i.e.* using (82) instead of (79) (see Section 4.2.1 for a detailed description of the method). The quantity of interest is thus:

$$\underline{f} = \underline{\mathrm{div}}\left(\underline{\underline{\mathrm{grad}}}\ \underline{X} + \left(\underline{\underline{\mathrm{grad}}}\ \underline{X}\right)^T\right) \tag{100}$$

This is usually computed from the velocity field, integrating the strain rate tensor $\underline{\underline{\dot{\gamma}}}$ and using (79):

$$\underline{f}_a^U = \underline{D}_a\{2\,\underline{\underline{\dot{\gamma}}}^n + \Delta t(\underline{\underline{G}}_b\{\underline{u}_c\} + (\underline{\underline{G}}_b\{\underline{u}_c\})^T)^n\} \tag{101}$$

The proposed alternative consists in calculating $\underline{f}$ from the displacement field $\underline{X}$ using Espanol and Revenga's (2003) second order differential operator (20):

$$\underline{f}^X{}_a = \underline{L}_a^E\{1;\underline{X}_b\} \tag{102}$$

In order to compare the two approaches accuracy, we consider the periodic case of 2D Taylor–Green vortices for which the analytical velocity field is known:

$$\underline{u}(x,y,t) = \begin{pmatrix} -\cos\left(\frac{2\pi x}{L}\right)\sin\left(\frac{2\pi y}{L}\right)\\ \sin\left(\frac{2\pi x}{L}\right)\cos\left(\frac{2\pi y}{L}\right)\end{pmatrix}\exp\left(\frac{-2\eta t}{\rho}\right) \tag{103}$$

In order to quantify the error due to the SPH operators only, we consider a Cartesian grid of fixed SPH particles. Their velocity and corresponding theoretical displacement are imposed according to (103). The simulation is carried out for $L = 1, \eta/\rho = 0.1\ \mathrm{m}^2\cdot\mathrm{s}^{-2}$ and $L/\delta r = 100$. After 10 s of physical time, expressions (101) and (102) are evaluated and compared to the exact value (100) obtained with Mathematica (Wolfram Research, Inc., 2017). The comparison is performed calculating the error defined by:

$$E_a = \sqrt{\frac{\left(\underline{f}_a^{SPH} - f(x_a, y_a)\right)^2}{\frac{1}{N}\sum_{a\in\mathscr{F}} f(x_a, y_a))^2}} \tag{104}$$

where $\underline{f}_a^{SPH}$ is the quantity $\underline{f}$ calculated at particle a with SPH using either (101) or (102), $f(x_a, y_a)$ is the theoretical values calculated with Mathematica at the particle a position (x_a, y_a), and N is the total number of SPH particles. Results are plotted in Fig. 5 that shows the error we got for both approaches. We clearly see that the error is much larger using (79), with a maximum error twice the one obtained using (82). This demonstrates the accuracy gain due to the use of a second order operator instead of two iterative uses of a first order operator within the scope of collocated methods.

5. Simulation of 2D soil collapse

To validate the present model, a simulation of an experimental 2-D collapse of dry soil is first carried out. The experiment was conducted by Bui et al. (2008) and the experimental set-up is illustrated in Fig. 6. The material is composed of aluminium bars of diameter 1 mm and 1.5 mm, length 50 mm and density 2650 $\mathrm{kg\cdot m^{-3}}$. The material porosity is not specified by the authors so we approximate it using the highest compact binary circle packing, leading to $\phi \approx 0.1$. All other physical parameters were determined experimentally by Bui et al. (2008) and are summarized in Table 2.

The simulations were performed using approximately 20,000 particles of soil and 1400 vertex and boundary elements, with an initial particle spacing of $\delta r = 0.001$ m. In all simulations, the ratio between δr and the smoothing length h is $\delta r/h = 2$. The numerical speed of sound was set to 10 $\mathrm{m\cdot s^{-1}}$.

Fig. 7 shows the simulation result at four physical times. We see that (49) leads to an accurate calculation of the yield stress, free of the typical SPH numerical oscillations of pressure. Besides, almost no deformation occurs between $t = 1.5$ s and $t = 2.0$ s, thus demonstrating that the model is able to represent the solid state even after a large time compared to the characteristic time of the experiment $\sqrt{H/g} = 0.1$ s. In Fig. 8, the surface configuration at the end of the experiment, as well as the failure line, are compared with numerical results obtained at 1.5 s. Experimentally, the failure line is defined as the line delimiting the non-deformed region. Numerically, we define the non-deformed region as the area where the displacement magnitude is less than the grain size, *i.e.* 1.5 mm. Both surface configuration and failure line are well predicted by the model although we assume that the granular medium is incompressible while it is not a valid hypothesis for dry soils. Thus, compressible effects are probably negligible in that case.

Note that we observed significant numerical noise when the solid state is reached. When its velocity tends to zero, the material should reach a steady motionless state and behave like a pure elastic solid. In the present model, we thus expect the shear forces to be solely calculated from the elastic solid model, *i.e.* the blending function should be homogeneously equal to zero within the material. However, because of numerical noise, some strain rate is artificially produced. Consequently, the sediment is in an unstable equilibrium, continuously oscillating between a solid state and a highly viscous liquid state. This emphasizes the fact that the blending approach (70) has a stabilization effect: contrary to Bui et al. (2008), the tensile instability is not directly observed. However, it still leads to numerical noise when the material is in the solid state. To circumvent this issue, the artificial stress method (Gray et al., 2001) could be tested in a future work. This technique was developed to remove the tensile instability that may be partly responsible for the above mentioned numerical noise.

6. Simulation of 2D dam-break wave on movable beds

The present model was also assessed to a case of dam-break wave propagating over a saturated granular bed. This study is based on the small-scale experiments carried out by Spinewine and Zech (2007). The experiments were performed in a flume being 25 cm wide and 6 m long. The initial configuration of the experiment is illustrated in Fig. 9. The dam-break is initiated by the sudden lowering of a gate that is not taken into account in the simulation. Two materials (sand and PVC) are tested to investigate the capability of the model to exhibit different behaviours with respect to different materials. The physical parameters of the materials are summarized in Table 3. Note that experimental Young's modulus and Poisson's coefficient were available for these materials. Consequently, we used generic values from Ulrich (2013) for similar cases. The first simulations were carried out setting a particle spacing of $\delta r = 0.002$ m, resulting in 525,000 particles of water, 150,000 particles of saturated soil and 13,000 particles vertex and boundary particles. The numerical speed of sound and isentropic coefficient are set identically for the saturated granular material and the water, that is $c_0 = 37\ \mathrm{m\cdot s^{-1}}$ and $\xi = 7$. Rhie and Chow's (1983) chequerboard correction (27) was applied with a coefficient of $\Lambda_{RC} = 1.0$, and no background pressure was used.

Fig. 10 shows a comparison of the experiment and the SPH simulation carried out for the sand for $\delta r = 2d_g$. We can see that the dynamics of the flow is qualitatively well reproduced by the model.

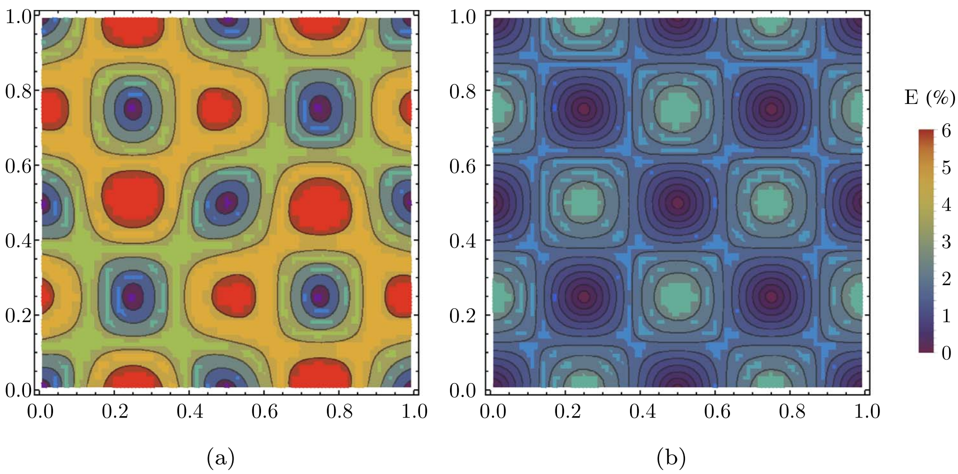

Fig. 5. Taylor–Green vortices – error on second order derivative of the displacement field in (a) calculated from strain rate tensor integration with (79) and in (b) calculated from displacement with (82). Results obtained for $L/\delta r = 100$ m and $\eta/\rho = 0.1$ m$^2\cdot$s^{-2} after 10 s of physical time.

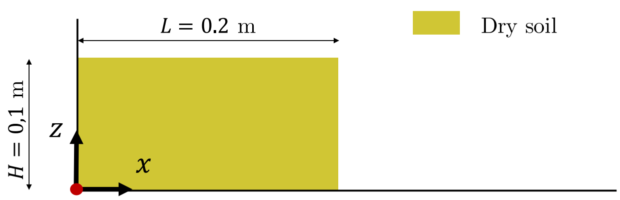

Fig. 6. 2D soil collapse – experimental set-up from Bui et al.'s (2008) experiments.

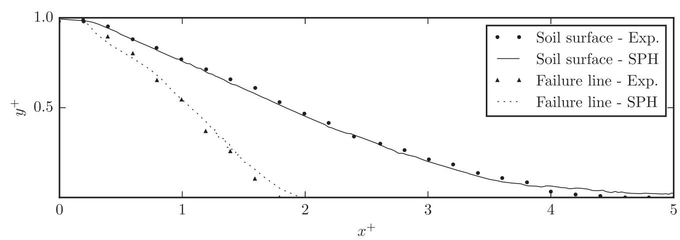

Fig. 8. 2D soil collapse – surface configuration and failure line at $t = 1.5$ s. Comparison of the present model to Bui et al.'s (2008) experiments.

Table 2
2D soil collapse – physical parameters of the granular materials (aluminium bars).

	ρ_g [kg/m^3]	d_g [mm]	ϕ [1]	E [Pa]	ν [1]	ψ [°]
Al.	2650	1.0 − 1.5	0.10	$8.4\cdot10^5$	0.3	19.6

Total eroded mass of soil –. First, let us focus on the total mass of sediments that reaches the downstream flume outlet. To do so, we define the relative error on the total eroded mass with respect to the experimental value:

$$Er = \frac{m_{\mathrm{SPH}} - m_{\mathrm{exp}}}{m_{\mathrm{exp}}} \tag{105}$$

Fig. 11 shows the error Er as a function of the ratio $\delta r/d_g$ (particle size/grain diameter). We can see that sand and PVC simulations exhibit similar behaviours although they have different grain sizes:

$$d_g^{\mathrm{PVC}} \approx 2\, d_g^{\mathrm{Sand}} \tag{106}$$

For the two materials, decreasing the SPH particle size improves the result until the critical value $\delta r = d_g$. Below this threshold, the soil model is no more valid and erosion is dramatically overestimated. This discretization limitation may be due to the fact that SPH particles are

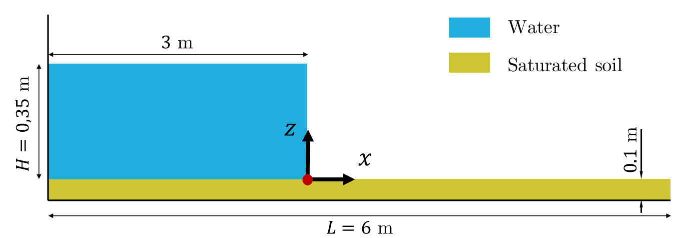

Fig. 9. Dam-break wave on movable beds – experimental set-up of Spinewine and Zech's (2007) experiments.

Table 3
Dam-break wave on movable beds – physical parameters of the granular materials.

	ρ_g [kg/m^3]	d_g [mm]	ϕ [1]	E [Pa]	ν [1]	ψ [°]
Sand	2683	1.89	0.47	$8.0\cdot10^5$	0.3	30
PVC	1580	3.90	0.42	$8.0\cdot10^5$	0.3	38

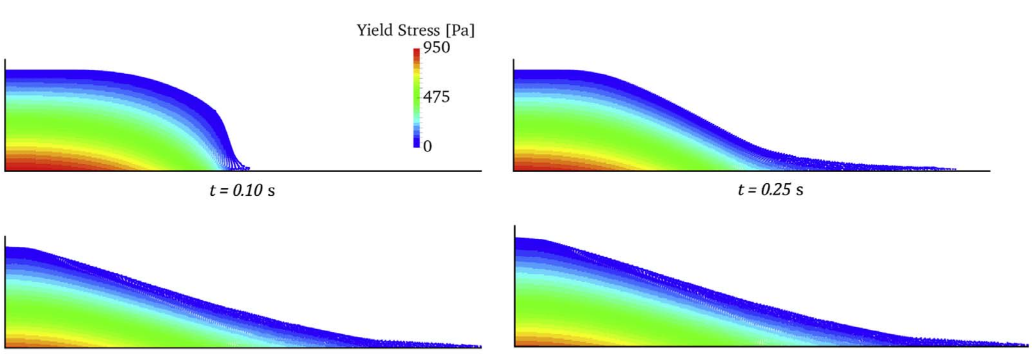

Fig. 7. 2D soil collapse – simulation results at $t = 0.1$ s, $t = 0.25$ s, $t = 1.5$ s and 2.0 s.

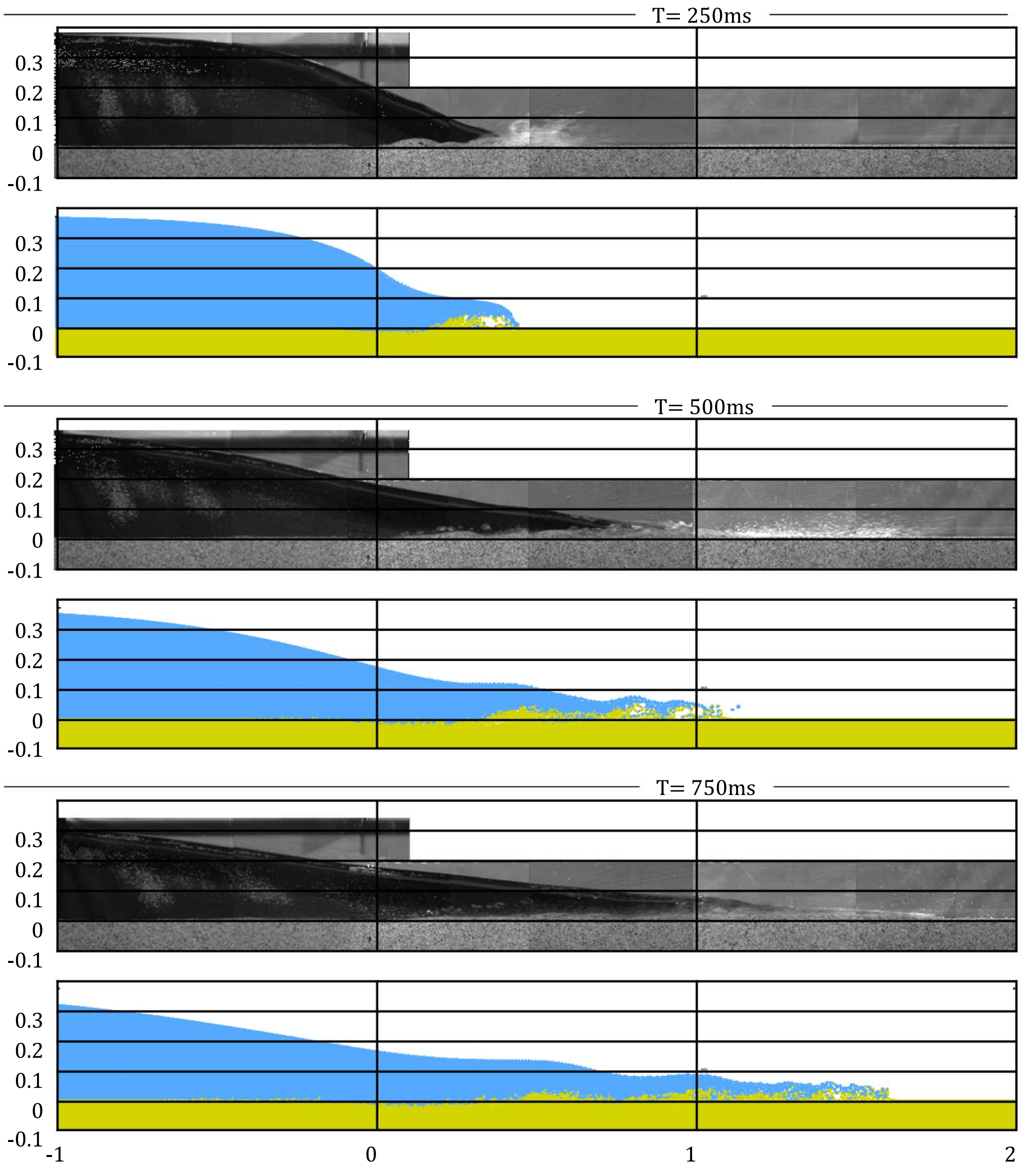

Fig. 10. Dam-break wave on movable beds – comparison of experimental and numerical results, with sand at time t = 250 ms, t = 500 ms and 750 ms, for $\delta r = 2d_g$.

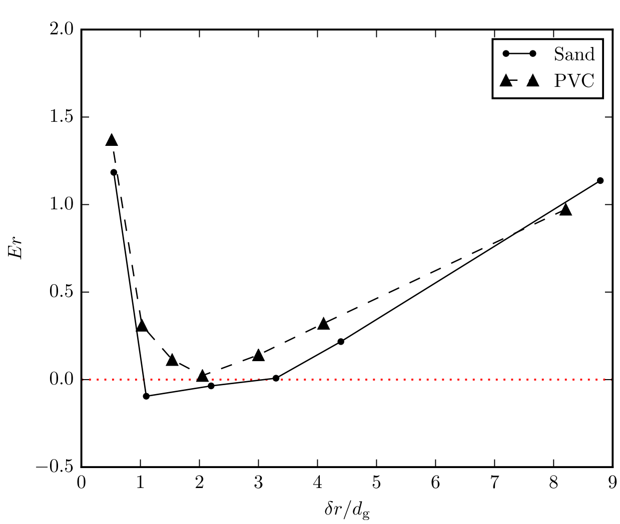

Fig. 11. Dam-break wave on movable beds – error on eroded mass as a function of the particle size/grain diameter ratio.

supposed to represent a macroscopic volume of the saturated soil.

The continuum hypothesis amounts to assume that the physical state of a heterogeneous system (*e.g.* a multi-phase system involving fluid and solid discrete particles) can be described by continuous field variables, whose space and time dependence is represented by differential equations for mass, momentum and energy balances (Baveye and Sposito, 1984). Such a description is based on the concept of representative elementary volume over which a local volume average allows "*a transition from microscopic physical properties to macroscopic field variables as well as from microscopic differential balance laws to macroscopic differential balance laws*" (Baveye and Sposito, 1984). The continuous fields result from the microscopic properties of the system, but they are macroscopic quantities that do not have any physical sense at the microscopic scale. Thus the representative elementary volume should be small enough (with regards to the problem dimensions) to be an approximation of a mathematical neighbourhood of its centre, but large enough to enclose many spatial heterogeneity (Baveye and Sposito, 1984). As an example, the derivation of Darcy's equations, that describes flows in porous media, requires that the characteristic length of the pores is smaller than the characteristic length of the averaging

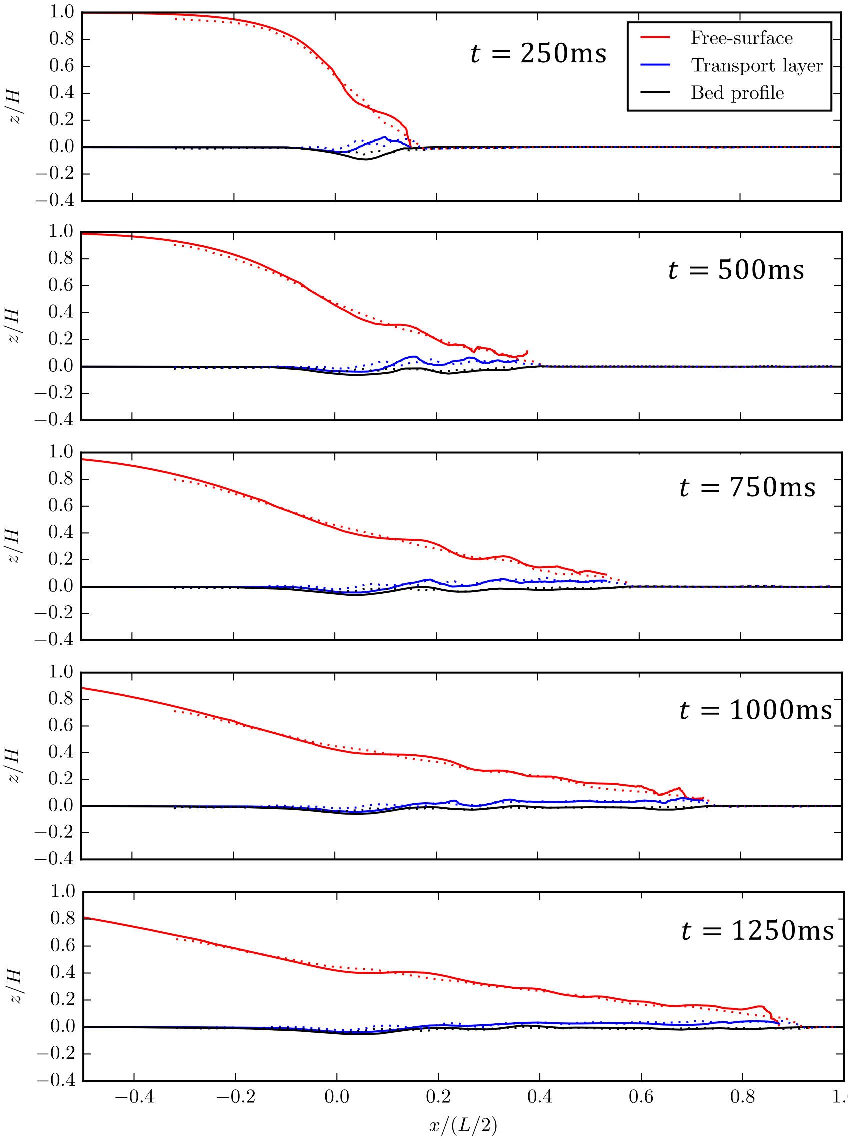

Fig. 12. Dam-break wave on movable beds – comparison of experimental (dotted lines) and numerical results (solid lines), with sand at time $t = 250$ ms, $t = 500$ ms, $t = 750$ ms, $t = 1000$ ms and $t = 1250$ ms, for $\delta r = 2d_g$. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

volume (Whitaker, 1986). Using SPH particles or mesh cells smaller than the pores (or the grains in our case) amounts to using averaging volumes that do not satisfy this constraint. Thus, the numerical solution approaches a theoretical solution that is not a correct description of the physical problem. Issues related to the length scales constraints and the continuum hypothesis have been addressed in the frame of porous media flows (Baveye and Sposito, 1984; Whitaker, 1986). In the scope of multi-phase flows, Gómez and Milioli (2005) claim that "*a minimum limit should be enforced on numerical spatial mesh sizes having in view the validity of the average Eulerian continuum equations for the solid phase*". According to Cleminș (1988), using mesh cells whose size does not satisfy the minimum size condition may lead to non-physical fluctuations of the resolved fields. In that case it is necessary to average the result on a larger volume to get relevant physical quantities. Extending this conclusion to Lagrangian methods seems to raise additional issues. Indeed, if a non-physical fluctuation of the velocity field occurs within an SPH particle, the particle moves and the simulation is irreversibly affected. Contrary to mesh-based methods, it is not possible to average the fluctuating fields afterwards to get relevant physical quantities.

Therefore, an SPH particle should not be smaller than the sediment grain size, otherwise the model would lead to deformations that do not occur in the real material, and overestimated strain rates. This may be partly due to the fact that the macroscopic elastic properties (*e.g.* Young's modulus) of the granular materials are very different from elastic properties of individual grains they are composed of Guan et al. (2012). Consequently, a particular attention must be paid to the discretization limitations when modelling granular materials with SPH. This might be avoided using alternative rheological laws or different elastic properties when $\delta r < \; < d_g$. On the other hand, Fig. 11 clearly shows that a sub-particle model would be necessary to correctly predict erosion when $\delta r > \; > d_g$. Within the scope of the present elastic-viscoplastic model, the problem depends on 10 physical parameters: the density ρ_w and dynamic viscosity of water η_w, the grain density ρ_g, the granular material porosity ϕ, the internal friction angle ψ, the Young's modulus E, the Poisson's coefficient, the grain size d_g, the gravity acceleration g and a characteristic length of the problem L. Applying the Buckingham π theorem, we obtain 7 dimensionless parameters that characterize this problem:

$$\phi, \quad \frac{\rho_g}{\rho_w}, \quad \psi, \quad \nu, \quad Re = \frac{g^{1/2}L^{3/2}\rho_w}{\eta_w}, \quad \frac{d_g}{L}, \quad \frac{E}{d_g(\rho_g - \rho_w)g} \tag{107}$$

In a way, using $\delta r < d_g$ amounts to setting $d_g' \approx \delta r$. We would then have SPH particles being solid matter while other are viscoplastic. The latter two dimensionless numbers are thus modified as:

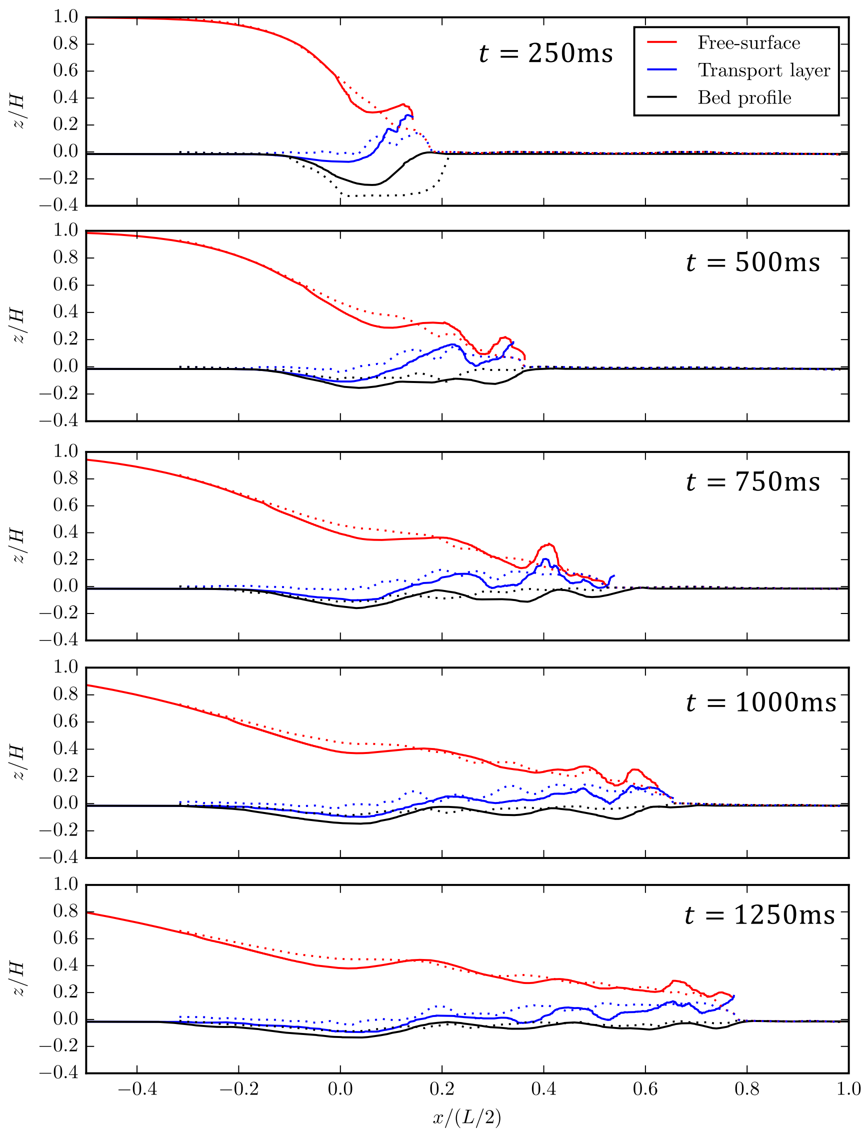

Fig. 13. Dam-break wave on movable beds – comparison of experimental (dotted lines) and numerical results (solid lines), with PVC at time $t = 250$ ms, $t = 500$ ms, $t = 750$ ms, $t = 1000$ ms and $t = 1250$ ms, for $\delta r = 2d_g$.

$$\frac{\delta r}{L},\quad \frac{E}{\delta r(\rho_g - \rho_w)g} \tag{108}$$

Regarding the first one, the decrease of the grain size has probably no influence on the results since we have $\delta r/L << 1$ as well as $d_g/L << 1$. However we see that the second dimensionless number is significantly changed. Recovering its physical value can be done by changing Young's modulus: when $\delta r = \alpha d_g$ with $\alpha < 1$, we would set the Young's modulus to αE. However, this was not tested in the scope of this work.

In Fig. 11 we see that an optimum seems to be achieved for $\delta r \in [d_g; 3d_g]$ for both sand and PVC. However, it must be pointed out that $\delta r > 3d_g$ is a coarse spatial discretization for this problem (≈ 22 particles on the water height). Thus, the error due to the hydrodynamics solver is also probably partly responsible on the erosion overestimation. Note that erosion is overestimated for almost all configurations. Thus, at worst, simulations give a superior limit of the real erosion.

In the following, results are presented for $\delta r \approx 2d_g$, *e.g.* for the sand, simulations were carried out setting a particle spacing of $\delta r = 0.004$ m, resulting in 130,000 particles of water, 38,000 particles of saturated soil and 3000 particles vertex and boundary particles.

Flows interfaces – . Fig. 12 shows the interface between moving and motionless sediment (black), the water-sediment interface (blue) and the water free-surface (red) for the sand. The interface between the motionless and the moving sediment is defined in line with the experimental detection method. In the experiment, a camera set to 200 images per second (*i.e.* one image every $\Delta t = 5$ ms) was used to acquire image sequences of the flow. Every image was then subtracted to the preceding image. The black region in the resulting image was defined as the motionless region. Given that the spatial resolution for one pixel is approximately $\Delta x = 1$ mm (Spinewine and Zech, 2007), we define a threshold velocity below which the sediment SPH particle is considered as motionless:

$$u_{\text{motion}} = \frac{\Delta x}{\Delta t} \tag{109}$$

Continuous lines represent the numerical results while experimental data are represented as dotted lines. A general good agreement is obtained for the sand bed. Interfaces as well as free-surface and water front are well predicted for the five considered times. Fig. 14 shows a comparison of flow interfaces for $\delta r = 2d_g$ (solid lines) and $\delta r = d_g/2$ (dashed lines). We clearly see that having SPH particles smaller that the grain size (dashed lines) leads to overestimated bed deformations while the results obtained for $\delta r = 2d_g$ (Figs. 12 and 14) match the experimental data.

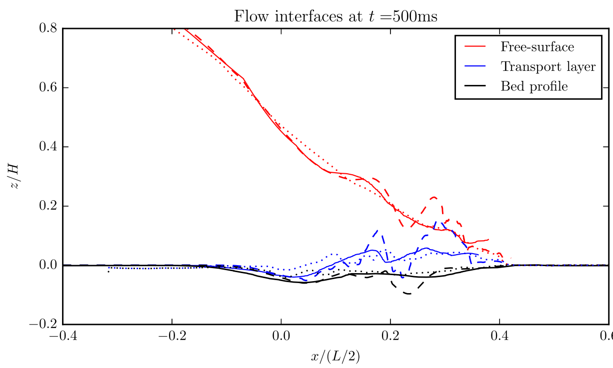

Fig. 14. Dam-break wave on movable beds – comparison of experimental (dotted lines) and numerical results, with sand at time $t = 500$ ms, for $\delta r = d_g/2$ (dashed lines) and $\delta r = 2d_g$ (solid lines).

For the PVC bed, Fig. 13 shows that the initial liquefaction is slightly underestimated. In particular, the experimental bed profile (*i.e.* the black dotted line) at $t = 250$ ms is deeper than the numerical one. This may be due to the fact that the lowering of a gate is not taken into account in the simulation. Moreover, there is a large uncertainty on PVC beds elastic properties. Indeed, elastic properties of sand are widely used in soil mechanics and a lot of experimental data is available. We did not find such data for PVC soils, hence the differences between numerical and experimental results. Nevertheless, simulations give good results for the PVC too. In addition, they exhibit very different behaviours for sand and PVC, demonstrating the capability of the model to account for different material properties.

7. Conclusion

In this work, Ulrich's (2013) elastic-viscoplastic model was implemented and improves in the frame of a SPH multi-phase formulation. Hu and Adams's (2006) multi-phase formulation was adapted to USAW boundary conditions and Vila's (1999) continuity equation is used so that the model can handle free-surface flows. The soil is treated as a continuum whose behaviour depends on a yield stress determined according to Drucker–Prager's criterion. In unyielded regions, the shear stress is calculated in line with linear elastic theory. In yielded regions, a shear thinning rheological law is used and the transitions between solid and liquid states are ensured by a blending function driven by the strain rate magnitude. A yield strain rate definition, based on the yield stress and physical properties of the soil, was proposed. Thus, contrary to Ulrich (2013), the mechanical behaviour of the soil does not depend on any numerical parameter. In addition, a reliable method, based on a Laplace equation, was developed to compute the effective pressure that is mandatory to compute the yield stress.

This elastic-viscoplastic model was tested on a two-dimensional soil collapse test case. The soil surface configuration and the failure line were compared to experimental results and a good agreement was found, the surface remaining constant after a few seconds. The model was then applied to a two-dimensional dam-break wave on movable beds. Two different materials were tested, *i.e.* sand and PVC pullets. Numerical free-surface, moving-soil/water and motionless-soil/moving-soil interfaces were compared to experimental data. A good agreement was found for the sand. For the PVC, the lack of elastic properties experimental values leads to less accurate but still reasonable results. For both materials, the total mass of eroded sediment is predicted with a correct order of magnitude when the SPH particle size does not exceed three times the grain size. Note that this work may be implemented in future versions of the open source GPUSPH code (GPUSPH official website).

To improve results, turbulence models could be tested. Indeed, the force exerted by water on the sediment is strongly related to the turbulent viscosity of water near the soil-water interface. Moreover, a sub-particle model could improve erosion prediction when the particle size is large compared to the grain size. More radically, unified solid-liquid models (Peshkov and Romenski, 2016) could be considered to model granular flows in SPH.

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